ORIUS: A Universal Runtime Safety Layer for Physical AI under Degraded Observation — Primary Battery Validation with Bounded AV and Healthcare Evidence

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***Abstract*—Physical AI systems operating in safety-critical domains face a fundamental hazard: *degraded observation*. When telemetry is delayed, dropped, stale, or drifted, a controller may evaluate its action as safe with respect to the observed state while the true physical state has already crossed a constraint boundary. We call this the *Observation–Action Safety Gap* (OASG).**

We present ORIUS (Observation–Reality Integrity for Universal Safety), a typed runtime layer that wraps any upstream controller and enforces safety through a five-stage kernel: Detect degraded telemetry, Calibrate the observation-consistent state set, Constrain the admissible action set, Shield unsafe candidate actions via repair or fallback, and Certify every released action with an auditable safety certificate.

The ORIUS architecture is *universal by design*: the kernel, benchmark schema, contract stack, and governance surface are domain-agnostic. Domain specifics are isolated in thin adapter modules that implement six typed methods. This paper reports a three-domain validation program with one primary validation domain and two bounded cross-domain validation domains:

1) **Battery Energy Storage: dispatch optimisation under degraded SoC telemetry, validated on locked CPSBench replay artifacts. Battery provides the deepest theorem-to-code-to-artifact closure; the promoted runtime slice achieves zero true-state violations.**
2) **Autonomous Vehicles: ego-speed and relative-gap prediction under degraded perception, validated on the all-zip grouped nuPlan replay surface with 37,344 held-out runtime scenarios, 1,531,104 replay steps, 1,531,104 certificates, empirical ORIUS *replay* TSVR 0.000163 *(replay surface; not closed-loop road deployment)*, and explicit non-closure of road deployment or full autonomous-driving field closure.**
3) **Medical and Healthcare Monitoring: bounded alert-release monitoring under degraded vital-sign observations, validated through a MIMIC-backed bounded runtime and governance surface.**

All promoted runtime surfaces maintain 100% certificate-chain validity, 100% audit completeness, and zero contract failures on their active runtime surfaces. All results are reproducible from locked artifacts with SHA-256 hashes. We report a strict theorem core, the complete shift-aware adaptive uncertainty mechanism, and a three-domain synthesis showing that the same kernel survives domain change without semantic loss while keeping battery-specific proof depth clearly distinguished from the bounded AV and Healthcare evidence.

***Index Terms*—runtime safety, degraded observation, conformal prediction, safety certificates, cyber-physical systems, battery energy storage, autonomous vehicles, healthcare monitoring, OASG**

I. Introduction

The deployment of AI-driven controllers in safety-critical physical systems—battery energy storage, autonomous vehicles, and clinical monitoring—has progressed from laboratory prototypes to field-deployed assets. A pervasive but under-addressed hazard accompanies this transition: the controller takes actions conditioned on *observations* of the true physical state, yet observations are never perfect. Sensors stall, telemetry packets arrive late or out-of-order, calibration drifts silently, and network links drop entire channels. When the observation pipeline degrades, the gap between what the controller *believes* to be the state and what the state *actually is* may grow unboundedly, and the controller may take an action that is formally safe in the observed state space yet catastrophically unsafe in the true state space.

We call this gap the **Observation–Action Safety Gap** (OASG). The OASG is distinct from standard notions of model uncertainty, adversarial perturba-

tion, or out-of-distribution detection.[1] It requires a runtime-level mechanism that:

1) Detects telemetry degradation at every control step.
2) Quantifies the resulting *set of states consistent with the degraded observation*.
3) Tightens the admissible action space so that safety is guaranteed for every member of that set.
4) Shields or repairs any candidate action that exits the tightened set.
5) Issues an auditable, tamper-evident certificate for every released action.

No existing framework—simplex architectures [1], runtime-verification monitors [2], Bayesian OOD detectors [3], or conformal-prediction wrappers [4]—addresses all five requirements simultaneously while remaining *domain-agnostic*, *i.e.* applicable to any cyber-physical domain via a thin adapter rather than a domain-specific redesign.

A. The ORIUS Framework

This paper presents **ORIUS** (Observation–Reality Integrity for Universal Safety), a typed runtime safety layer whose five-stage kernel—**Detect**, **Calibrate**, **Constrain**, **Shield**, **Certify**, abbreviated DC3S—closes the OASG for any upstream controller. The kernel is defined over abstract typed contracts (Section V); domain specifics are confined to an adapter module that implements six typed methods. A formal theorem chain (T1–T11) connects the runtime invariants to end-to-end safety guarantees under stated assumptions (A1–A13).

B. Scope and Claims

Unlike broader planning drafts that motivated ORIUS with additional candidate domains, this manuscript restricts its empirical claim surface to **three active domains** with locked, reproducible artifacts and explicit evidence tiers:

1) **Battery Energy Storage (BESS)**: dispatch optimisation under degraded state-of-charge telemetry on locked CPSBench replay artifacts. Battery is the primary validation domain and carries the deepest artifact-to-proof closure.
2) **Autonomous Vehicles (AV)**: longitudinal prediction of ego-speed and relative gap under degraded perception, validated on the completed all-zip grouped nuPlan replay surface with explicit non-closure of full autonomous-driving field deployment.
3) **Medical and Healthcare Monitoring**: bounded monitoring and alert release under degraded vital-sign observations. Healthcare is validated as a bounded MIMIC-backed runtime domain; its claims remain narrower than the battery domain.

The three active domains share the same kernel binary, the same contract stack, the same governance pipeline, and the same CertOS certificate chain.

C. Contributions

The principal contributions of this paper are:

1) A complete formal architecture for runtime safety under degraded observation, expressed as five typed stages with a supporting contract algebra (Section IV).
2) An 11-theorem proof chain connecting observation reliability to true-state safety guarantees (Section VI).
3) A shift-aware adaptive uncertainty mechanism that dynamically adjusts prediction intervals without offline recalibration (Section VII).
4) A bounded three-domain empirical lane:
   - **Battery primary validation domain**: zero true-state violations on the promoted runtime slice, with the deepest theorem-to-artifact alignment.
   - **AV bounded validation domain**: 1,531,104 all-zip grouped nuPlan replay steps and certificates; ORIUS reduces TSVR from 0.289250 for the no-wrapper baseline to 0.000163 on the promoted replay row, while the harder relative-gap shift table remains explicitly marked as bounded.
   - **Healthcare bounded validation domain**: TSVR reduced from 0.194489 to 0.000000 on the promoted MIMIC monitoring surface, with grouped calibration and governance evidence reported explicitly as bounded claims.
5) A fully reproducible artifact bundle with SHA-256 hashes, locked dependencies, and a single-command pipeline (Section XI).

a) What this paper proves and what it does not.: The theorem chain (T1–T11) is stated on the typed-contract algebra, but the proof posture is tracked theorem-by-theorem in the active theorem-evidence matrix rather than flattened to "proved in full generality." Empirical confirmation is restricted to the three domains above. In particular:

- The Battery domain achieves TSVR = 0 (zero true-state violations) and serves as the *pri-*

[1]Out-of-distribution (OOD) detectors identify distributional mismatches but do not synthesise conservative action sets; adversarial robustness certifies bounded input perturbations but does not handle telemetry infrastructure faults (packet loss, reordering, drift).

mary validation domain: it demonstrates the full pipeline end-to-end with the deepest artifact-to-theorem closure.
- The AV domain is a *bounded validation* domain: ORIUS is evaluated only under the longitudinal TTC + predictive-entry-barrier contract, with bounded nuPlan replay artifacts and no claim of full road deployment closure.
- The Healthcare domain is a *bounded validation* domain: ORIUS is evaluated only for monitoring and alert-release semantics on the MIMIC-backed healthcare lane. Certificate-valid release is reported from runtime-emitted healthcare CVA on this surface.
- Shift-awareness guarantees are bounded to the drift rates observed during validation; adversarial robustness under intentional telemetry attacks has not been evaluated.

D. Paper Organisation

Section II formalises the OASG problem. Section III surveys related work. Section IV details the DC3S kernel and typed interfaces. Section V covers the contract stack and governance surface. Section VI presents the theorem bridge connecting contracts to safety. Section VII introduces the shift-aware mechanism. Section VIII and Section IX provide the deepest per-domain empirical sections for BESS and AV, while Section X synthesises the active three-domain lane, including the bounded Healthcare domain. Section XI documents the reproducibility infrastructure. Section XII discusses limitations. Section XIII concludes.

II. Problem Formulation: The Observation–Action Safety Gap

A. System Model

Consider a discrete-time cyber-physical system (CPS) with true state $x_t \in \mathcal{X} \subseteq \mathbb{R}^n$ evolving under dynamics

$$x_{t+1} = f(x_t, a_t, \omega_t), \tag{1}$$

where $a_t \in \mathcal{A}$ is the control action and ω_t is an exogenous disturbance. The controller does not observe x_t directly. Instead it receives a *telemetry observation*

$$z_t = h(x_t) + \epsilon_t, \tag{2}$$

where $h : \mathcal{X} \to \mathcal{Z}$ is the sensor model and ϵ_t is measurement noise. Under nominal conditions, the safety literature assumes ϵ_t is bounded by some known envelope $\bar{\epsilon}$.

TABLE I: Fault families used to instantiate degraded observation.

Fault family	Description
`dropout`	Complete packet loss for one or more channels
`stale_sensor`	Sensor value frozen at its last known value
`delay_jitter`	Timestamp reordering or variable latency
`drift_combo`	Slow additive or multiplicative calibration drift
`spikes`	Transient high-magnitude outlier readings
`out_of_order`	Packet reordering violating causal ordering

B. Degraded Observation

In practice, the observation pipeline introduces additional corruption beyond ϵ_t. We model degraded observation as a map

$$\tilde{z}_t = g(z_t, \delta_t), \tag{3}$$

where δ_t is an *infrastructure fault vector* drawn from a fault family $\mathcal{F}$. In our implementation, the fault family includes:

The degraded observation $\tilde{z}_t$ may differ from z_t by an arbitrarily large amount that depends on the fault severity and duration.

C. Formal Definition of the OASG

Definition II.1 (Observation-Consistent State Set)**.** Given a degraded observation $\tilde{z}_t$ and reliability weight $w_t \in [0, 1]$, the *observation-consistent state set* is

$$U_t(\tilde{z}_t, w_t) = \Big\{ x \in \mathcal{X} : \big\| h(x) - \tilde{z}_t \big\| \le r(w_t) \Big\}, \tag{4}$$

where $r : [0, 1] \to \mathbb{R}_+$ is a monotonically non-increasing function satisfying $r(1) = \bar{\epsilon}$ (nominal radius) and $r(w) \to \infty$ as $w \to 0$ (complete loss of trust).

In the ORIUS implementation, $r(w_t)$ is realised by conformal prediction intervals inflated by the *planned contract factor*

$$\alpha = \mathrm{clip}\big(1.0 + 0.7\,(1 - w_t) + 0.5\,s_{\text{shift}},\ 1.0,\ 3.0\big), \tag{5}$$

where s_{shift} is a shift score computed by the adaptive mechanism (Section VII).

Definition II.2 (Observation–Action Safety Gap)**.** Let $S \subset \mathcal{X}$ be the safe region and let π be a controller policy. The *OASG* at time t is the event

$$\begin{aligned} \mathrm{OASG}_t = \big\{ \pi(\tilde{z}_t) \in C(\tilde{z}_t) \big\} \\ \wedge \big\{ \exists\, x \in U_t : f(x, \pi(\tilde{z}_t), \omega_t) \notin S \big\}. \end{aligned} \tag{6}$$

The OASG is non-empty whenever there exists a true state consistent with the degraded observation for which the controller's action leads to a safety violation.

D. Closing the Gap: Requirements

Closing the OASG requires the runtime layer to satisfy five requirements:

- **R1 Detection**: At every step, assess the reliability w_t of the current observation using both statistical detectors and history-based heuristics.
- **R2 Calibration**: Compute U_t as a conservative superset of the true-state-consistent region.
- **R3 Constraint Tightening**: Derive a restricted action set $\mathcal{A}_t \subseteq \mathcal{A}$ such that every action in $\mathcal{A}_t$ is safe for *all* states in U_t.
- **R4 Shielding**: If $a_t^* = \pi(\tilde{z}_t) \notin \mathcal{A}_t$, repair or replace a_t^* with $a_t^{\text{safe}} \in \mathcal{A}_t$.
- **R5 Certification**: Produce a safety certificate binding the released action, the observation, the reliability assessment, and the constraint check into an auditable record.

These five requirements map directly to the five stages of the DC3S kernel.

E. True-State Violation Rate (TSVR)

The primary safety metric is the *true-state violation rate*:

$$\text{TSVR} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\big[f(x_t, a_t^{\text{released}}, \omega_t) \notin S\big]. \quad (7)$$

A runtime layer with TSVR $= 0$ has zero true-state violations. In the battery domain, all four controller configurations achieve TSVR $= 0$. In the all-zip grouped nuPlan AV replay row, ORIUS reduces the TSVR from 0.289250 (baseline) to 0.000163 over 1,531,104 runtime steps.

F. Additional Metrics

Beyond TSVR, we track:

- **OASG rate**: fraction of steps where the safety gap is detected and closed.
- **Coverage Validity Assessment (CVA)**: fraction of steps where the conformal interval contains the true state.
- **Governance Digest Quality (GDQ)**: composite governance metric measuring certificate completeness and payload validity.
- **Intervention Rate (IR)**: fraction of steps where the shield modifies the candidate action.
- **Audit Completeness**: fraction of steps with a valid, non-expired certificate on record.

G. Empirical Witness for the OASG

Figure 1 provides the empirical witness for Definition II.2: under a quality-ignorant baseline, there exist time steps where the controller action is safe with respect to the observed state but violates the true-state constraint. The Battery domain exhibits 20 such counterexamples; the all-zip grouped nuPlan AV baseline exhibits an OASG rate of 0.0230 over 1,531,104 steps.

III. Related Work

The ORIUS framework sits at the intersection of several research threads: runtime verification, conformal prediction, safety shielding, and CPS telemetry resilience. We survey each in turn, highlighting the gap that ORIUS addresses.

A. Runtime Verification and Monitoring

Runtime-verification (RV) monitors check whether an execution trace satisfies a temporal-logic specification [2, 5]. Classic RV monitors are *passive*: they flag violations but do not modify the controller's action. The simplex architecture [1] introduces a *decision module* that switches between a complex controller and a verified-safe baseline, but it assumes that the state is accurately known—the observation pipeline is outside its trust model.

ORIUS goes beyond passive monitoring by actively computing a constraint-tightened action set and repairing unsafe actions. Crucially, it operates on *degraded* observations and inflates the safety envelope accordingly, a capability absent from standard simplex or watchdog designs.

B. Conformal Prediction for Safety

Conformal prediction (CP) constructs distribution-free prediction intervals with finite-sample coverage guarantees [6]. Adaptive Conformal Inference [7] and related methods handle distribution shift online. Recent work applies CP to robotics and motion planning [8, 9].

ORIUS uses CP as the *calibration engine* (Stage 2 of DC3S) but wraps it in a full runtime safety layer with constraint tightening, action shielding, and governance certification. Stand-alone CP only produces intervals; ORIUS consumes them, inflates them according to degradation severity, and feeds the result into a formal contract stack.

C. Safety Shielding and Control Barrier Functions

Shield synthesis computes a reactive strategy that enforces a safety specification by overriding unsafe controller actions [10]. Control Barrier Functions (CBFs) [11] certify forward invariance of safe sets. Both approaches typically require a known, accurate state vector.

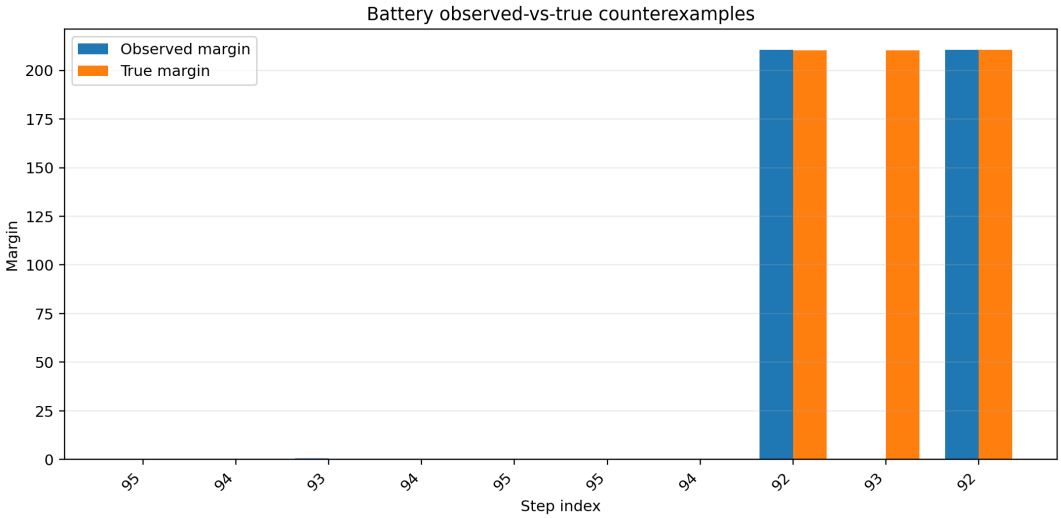


Fig. 1: Observed vs. true state counterexamples under a quality-ignorant baseline. Each point is a time step where the observed state satisfies the safety constraint but the true state does not—the defining signature of the OASG (Definition II.2). Battery domain shown; the bounded AV counterpart is reported in Section IX. Artifact: `reports/battery_av/battery/fig_observed_vs_true_counterexamples.png`.

ORIUS contributes by coupling the shield to an *observation-consistent state set* U_t rather than a point state estimate. The constraint-tightening stage (Section IV) is conceptually analogous to robust CBF formulations, but it is implemented via interval arithmetic on conformal sets, making it compatible with black-box learned models.

D. Telemetry Resilience in CPS

Fault-tolerant control and networked control systems address packet loss and delay [12]. However, these methods typically assume known fault models (*e.g.* Bernoulli packet loss with known rate) and linear dynamics. ORIUS's detect stage uses statistical detectors (Page–Hinkley, staleness counters) to *assess* the fault type without assuming a parametric model, and the calibration stage inflates uncertainty in proportion to the detected severity.

E. Comparison Summary

Table II summarises the coverage of requirements R1–R5 across related approaches.

TABLE II: Comparison of ORIUS with related approaches on Requirements R1–R5. ✓= addressed, (∼) = partial, × = not addressed.

Approach	R1	R2	R3	R4	R5
Classic RV monitor	∼	×	×	×	×
Simplex architecture	×	×	×	✓	×
Conformal prediction	×	✓	×	×	×
CBF / Shield synthesis	×	×	✓	✓	×
Fault-tolerant NCS	✓	∼	×	×	×
ORIUS (DC3S)	✓	✓	✓	✓	✓

F. Positioning

ORIUS is, to our knowledge, the first runtime layer that satisfies all five requirements simultaneously while remaining domain-agnostic through thin typed adapters. The defended three-domain lane we present combines a battery witness row with bounded AV and Healthcare rows, demonstrating that the same kernel—compiled once, configured via adapter—survives domain transfer without changing its runtime semantics.

IV. The DC3S Runtime Architecture

The ORIUS kernel orchestrates five typed stages, each consuming the output of the previous stage and producing a well-typed contract object. We describe each stage, its inputs and outputs, the typed contracts involved, and the algorithm that implements it.

A. Architecture Overview

B. Stage 1: Detect

The detect stage assesses the trustworthiness of the incoming telemetry.

a) Input.: An `ObservationPacket` containing the raw telemetry $\tilde{z}_t$, the parsed state dictionary, a UTC timestamp, and the observation history length.

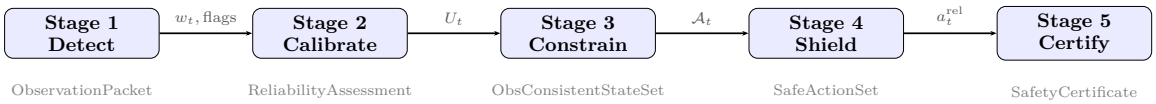


Fig. 2: The five stages of the DC3S kernel and their inter-stage typed contracts.

b) Algorithm.: Three detector families run in parallel:

1) **Staleness detector**: Checks whether the telemetry timestamp is within a configurable freshness window. If stale, a binary flag `is_stale` is raised and w_t is penalised.
2) **Page–Hinkley drift detector**: Maintains a running mean of each sensor channel and detects upward or downward drift using a cumulative-sum test with configurable threshold λ and tolerance δ [13, 14].
3) **History-based heuristics**: Computes the number of consecutive identical readings (freeze detection), the rate of out-of-order timestamps, and spike magnitude relative to a rolling window.

c) Output.: A `ReliabilityAssessment` containing:

- $w_t \in [0, 1]$: the composite reliability weight, clamped to $[0, 1]$.
- `flags`: a dictionary of boolean detector outputs.
- `drift_flag`: a boolean indicating whether drift was detected.
- `drift_meta`: metadata from the Page–Hinkley detector (cumulative sum, threshold, direction).
- `assumption_tags`: the assumptions activated by this assessment (A2, A5, A6).

C. Stage 2: Calibrate

The calibrate stage computes the observation-consistent state set U_t.

a) Input.: The parsed state, the uncertainty dictionary (conformal prediction intervals per target per horizon), and the reliability weight w_t.

b) Observation Quality Estimation (OQE).: The composite reliability weight is the product of three independent scores:

$$w_t = w_t^{\text{fresh}} \cdot w_t^{\text{range}} \cdot w_t^{\text{consist}}, \tag{8}$$

where:

- $w_t^{\text{fresh}} = \exp(-\lambda_f \cdot s_t)$, with λ_f the freshness decay rate and s_t the staleness counter (seconds since last novel reading).
- $w_t^{\text{range}} = \mathbf{1}[x_t \in [\underline{x}, \bar{x}]]$, the range-validity indicator.
- w_t^{consist} penalises readings that deviate from the physics-model prediction beyond a configurable tolerance.

c) Reliability-Adaptive Conformal (RAC) Inflation.: Two inflation forms appear in ORIUS and serve distinct roles, which we make explicit here to avoid conflation. The *theoretical envelope* that the risk-bound proofs (§VI) reference is the *RAC-Cert* form:

$$C_t^{\text{RAC}}(\alpha) \;=\; C_t(\alpha) \cdot \big[1 + \kappa_r\,(1 - w_t) + \kappa_d \cdot d_t + \kappa_s \cdot s_t\big], \tag{9}$$

where $C_t(\alpha)$ is the base conformal interval at miscoverage level α, $\kappa_r = 0.5$ is the reliability penalty, $\kappa_d = 0.3$ is the drift penalty, $d_t \in \{0, 1\}$ is the Page–Hinkley drift indicator, and $\kappa_s = 0.05$ is the staleness scaling with s_t the staleness count in seconds. Equation (9) is a closed-form upper envelope: it exposes each degradation source as a separate, additively monotone term, which is the form used to discharge Assumption A5 (Monotone Bounded Inflation) in the theorem chain.

The *runtime* executes a simplified, clipped form that collapses the drift and staleness terms into a single shift-aware score and bounds the total inflation between $1\times$ and $3\times$:

$$\boxed{\alpha = \text{clip}\big(1.0 + 0.7\,(1 - w_t) + 0.5\,s_{\text{shift}},\ 1.0,\ 3.0\big).} \tag{10}$$

Equation (10) is the only inflation function actually evaluated at runtime. Its implementation anchors are the AV runtime wrapper, the training feature builder, and the shift-aware uncertainty adapter; the exact file and line anchors are recorded in the theorem result cards and claim-evidence registers rather than repeated as long paths in the manuscript. The relationship between the two forms is that Eq. (10) is a pointwise lower bound of Eq. (9) once s_{shift} is identified with a conservative combination of $\kappa_d d_t + \kappa_s s_t$, so any safety guarantee proved under (9) continues to hold under the looser clipped form whenever Eq. (10) dominates the theoretical envelope, and the runtime is otherwise scoped by the explicit $\alpha \le 3$ ceiling.

d) Output.: An observation-consistent state set with:

- `observed_state`: the point estimate from the parsed telemetry.
- `lower_bounds`, `upper_bounds`: per-target dictionaries.
- `inflation`: the applied factor α.

- `quantile`: the conformal quantile level.
- `reliability_weight`: the w_t that produced this set.

D. Stage 3: Constrain

The constrain stage derives the safe action set $\mathcal{A}_t$ from U_t and the domain safety specification.

a) Input.: The Stage 2 uncertainty set and a `SafetySpec` encoding domain-specific constraints and the fallback action.

b) Algorithm.: For each constraint in the safety specification:

1) Evaluate the constraint predicate over the worst-case state within U_t.
2) If the constraint is violated under worst-case evaluation, mark it as binding.
3) The safe action set is the intersection of all non-binding constraint regions.

In the battery domain, constraints include SoC bounds ($\text{SOC}_{\min} \leq \text{SOC}_t \leq \text{SOC}_{\max}$) and power limits ($|P_t| \leq P_{\max}$). In the AV domain, constraints include minimum gap maintenance and ego-speed bounds.

c) Output.: A `SafeActionSet` containing the admissible region and the list of candidate actions with their tightened constraint evaluations.

E. Stage 4: Shield

The shield stage intercepts any candidate action that falls outside $\mathcal{A}_t$ and either repairs it or substitutes the domain-specific fallback.

a) Input.: The candidate action $a_t^* = \pi(\tilde{z}_t)$, the `SafeActionSet`, and the `SafetySpec` (which provides the fallback).

b) Algorithm.:

if $a_t^* \in \mathcal{A}_t$ **then**
  return a_t^* ▷ Action is safe; pass through
else
  Attempt projection: $a' = \text{proj}_{\mathcal{A}_t}(a_t^*)$
  if a' satisfies all constraints **then**
    return a' ▷ Repaired action
  else
    return a_t^{safe} ▷ Fallback action
  end if
end if

c) Output.: A `RepairDecision` containing the released action a_t^{rel}, whether it was passed through, repaired, or substituted, and the delta $\|a_t^* - a_t^{\text{rel}}\|$.

F. Stage 5: Certify

The certify stage produces a `SafetyCertificate` that binds together all previous stage outputs into an auditable record.

a) Output.: A `SafetyCertificate` containing:

- A unique certificate UUID.
- The released action and its provenance (pass-through, repair, fallback).
- The observation packet, reliability assessment, and uncertainty set.
- The constraint evaluation results.
- A timestamp and domain identifier.
- A payload dictionary for domain-specific metadata.

The certificate is appended to an append-only governance log. The `ContractVerifier` (Section V) validates every certificate against the contract stack before it is committed to the log.

G. Adapter Interface: The *DomainInstantiation* Protocol

Domain-specific behaviour is confined to an adapter that implements six typed methods defined by the `DomainInstantiation` protocol:

1) `ingest_telemetry(raw_bytes) -> ObservationPacket`: parse raw telemetry into a typed observation packet.
2) `compute_oqe(history) -> ReliabilityAssessment`: evaluate the OQE triple Equation (8) and produce the composite weight w_t.
3) `build_uncertainty_set(state, w_t, conformal_q) -> ObsConsistentStateSet`: inflate the base conformal interval using Equation (9).
4) `tighten_action_set(U_t, spec) -> SafeActionSet`: derive $\mathcal{A}_t$ via worst-case constraint evaluation over U_t.
5) `repair_action(a_star, A_t, spec) -> RepairDecision`: project or substitute the candidate action.
6) `emit_certificate(stages) -> SafetyCertificate`: bind all stage outputs into an auditable certificate.

The same DC3S kernel binary orchestrates both the battery adapter (`src/orius/battery_adapter.py`) and the AV adapter (`src/orius/av_adapter.py`). No kernel code is modified when switching domains.

H. The DC3S Kernel Algorithm

Algorithm 1 formalises the five-stage pipeline as a single procedural algorithm.

I. Battery-Domain Constraint Tightening: FTIT

In the battery domain, Stage 3 enforces the *Fault-Tolerant Invariant Tube* (FTIT), which propagates

Algorithm 1 DC3S Universal Kernel: `execute_universal_step`

Require: Domain adapter $\mathcal{D}$, raw telemetry $\mathbf{z}_t^{\text{raw}}$, candidate action a_t^*, safety spec $\mathcal{C}$, conformal quantile q_α, history buffer $\mathcal{H}$
Ensure: Released action a_t^{rel}, SafetyCertificate $\mathcal{K}_t$

[Stage 1: Detect]
$\text{obs}_t \leftarrow \mathcal{D}.\texttt{ingest_telemetry}(\mathbf{z}_t^{\text{raw}})$
$(w_t, \text{flags}, d_t, s_t) \leftarrow \mathcal{D}.\texttt{compute_oqe}(\mathcal{H} \cup \{\text{obs}_t\})$

[Stage 2: Calibrate]
$C_t(\alpha) \leftarrow \texttt{conformal_quantile}(q_\alpha, \text{obs}_t)$
$\alpha_t \leftarrow \text{clip}(1 + 0.7(1 - w_t) + 0.5\, s_{\text{shift}},\ 1, 3)$ ▷ Equation (10); runtime form, $\leq$ envelope (9)
$U_t \leftarrow [\text{obs}_t.\hat{x} - \alpha_t\, C_t(\alpha),\ \text{obs}_t.\hat{x} + \alpha_t\, C_t(\alpha)]$

[Stage 3: Constrain]
$\mathcal{A}_t \leftarrow \mathcal{D}.\texttt{tighten_action_set}(U_t, \mathcal{C})$

[Stage 4: Shield]
if $a_t^* \in \mathcal{A}_t$ **then**
  $a_t^{\text{rel}} \leftarrow a_t^*$ ▷ Pass-through
else
  $a' \leftarrow \text{proj}_{\mathcal{A}_t}(a_t^*)$ ▷ Repair attempt
  if $a' \in \mathcal{A}_t$ **then**
    $a_t^{\text{rel}} \leftarrow a'$
  else
    $a_t^{\text{rel}} \leftarrow a_{\text{safe}}$ ▷ Fallback
  end if
end if

[Stage 5: Certify]
$\mathcal{K}_t \leftarrow \mathcal{D}.\texttt{emit_certificate}(\text{obs}_t, w_t, U_t, \mathcal{A}_t, a_t^{\text{rel}})$
Append $\mathcal{K}_t$ to governance log
return $(a_t^{\text{rel}}, \mathcal{K}_t)$

the uncertainty from Stage 2 through the energy dynamics:

$$\bar{E}_{t+k} = \hat{E}_t + \delta_{\max} + k\,\eta^+\,P^+_{\max}\,\Delta t, \tag{11}$$

$$\underline{E}_{t+k} = \hat{E}_t - \delta_{\max} - k\,P^-_{\max}/\eta^-\,\Delta t. \tag{12}$$

The safe-action set for the battery is then:

$$\mathcal{A}_t = \{(P^+, P^-) \mid E_{\min} + q_t^{\text{RAC}} \leq \hat{E}_{t+1}(P^+, P^-) \leq E_{\max} - q_t^{\text{RAC}}\}, \tag{13}$$

where q_t^{RAC} is the RAC-inflated safety buffer derived from C_t^{RAC}.

J. Latency Profile

Table III reports the per-stage latency measured over 12,096 runtime steps (battery pipeline with 42 fault-injected scenarios).

TABLE III: DC3S per-stage latency profile (battery pipeline, 12,096 steps).

Stage	Mean	Med.	P95	Max
OQE (Detect)	0.026	0.024	0.030	0.210
Drift detect	0.000	0.000	0.000	0.003
Calibrate	0.012	0.011	0.015	0.180
Repair (Shield)	0.002	0.002	0.003	0.090
Full step	**0.042**	**0.037**	**0.041**	**2.416**

The mean full-step latency of 0.042 ms is well within the 10-minute dispatch interval for battery systems and the 100 ms control loop for autonomous vehicles. Even the P95 latency of 0.041 ms leaves ample headroom for real-time deployment.

K. Detailed Stage Walkthrough: Battery Example

To make the five-stage pipeline concrete, we trace the execution of a single battery dispatch step under a stale-sensor fault.

a) Step context.: The true SoC is 450 MWh. The sensor has been frozen for 3 intervals, so the telemetry reports `soc_mwh = 440` (10 MWh stale error). The promoted battery witness policy proposes a 50 MW discharge.

b) Stage 1 (Detect).: The staleness detector records 3 consecutive identical readings and raises `is_stale = true`. The reliability weight drops to $w_t = 0.976$. The Page–Hinkley detector does not fire because the constant readings mimic a zero-drift signal. Output: $w_t = 0.976$, `flags = {is_stale: true}`.

c) Stage 2 (Calibrate).: The battery runtime calibrator emits the base interval $\hat{C}_t = [420, 480]$ MWh (half-width 30 MWh). With $w_t = 0.976$ and $s_{\text{shift}} = 0$:

$$\alpha = \text{clip}\big(1.0 + 0.7 \times 0.024 + 0.5 \times 0,\ 1.0,\ 3.0\big) = 1.017.$$

The inflated interval is $\hat{C}_t^{\text{RAC}} = [419.5, 480.5]$ MWh. The true SoC of 450 MWh falls within $\hat{C}_t^{\text{RAC}}$, satisfying T1.

d) Stage 3 (Constrain).: The safety specification requires $\text{SOC}_{\min} = 100$ MWh $\leq$ SOC $\leq \text{SOC}_{\max} = 900$ MWh. Even under worst-case evaluation (SOC = 419.5 MWh), a 50 MW discharge for one interval (10 min) reduces SoC by ∼8.3 MWh to 411.2 MWh, still above $\text{SOC}_{\min}$. The action is admitted to $\mathcal{A}_t$.

e) Stage 4 (Shield).: Since the candidate action (50 MW discharge) is in $\mathcal{A}_t$, the shield passes it through without repair. Output: `pass_through`, delta = 0.

f) Stage 5 (Certify).: A certificate is emitted with UUID, the released action (50 MW discharge), the stale-sensor flag, $w_t = 0.976$, $\alpha = 1.017$, and the constraint evaluation result. The certificate passes all predicates CP1–CP6.

This walkthrough demonstrates that even moderate degradation (10 MWh stale error) is handled gracefully: the interval inflates by 1.7%, the constraint check passes, and no intervention is needed.

L. Detailed Stage Walkthrough: AV Example

We trace a single AV step under a spike fault.

a) Step context.: The ego vehicle is travelling at 12 m/s with a relative gap of 45 m. A spike fault injects a transient reading of $v_{\text{ego}} = 35$ m/s (23 m/s above truth).

b) Stage 1 (Detect).: The spike detector computes the z-score of 35 m/s against the rolling mean (11.5 m/s) and standard deviation (2.3 m/s): $z = (35 - 11.5)/2.3 = 10.2$. This far exceeds the spike threshold ($z > 3$), so `spike_detected = true` is raised and w_t drops to approximately 0.40.

c) Stage 2 (Calibrate).: With $w_t = 0.40$ and $s_{\text{shift}} = 0$:

$$\alpha = \text{clip}\big(1.0 + 0.7 \times 0.60 + 0.5 \times 0,\ 1.0,\ 3.0\big) = 1.42.$$

The base conformal interval for 1 s ego speed is $\hat{C}_t = [8.0, 16.0]$ m/s (half-width 4.0 m/s centred on 12 m/s). After inflation: $\hat{C}_t^{\text{RAC}} = [6.3, 17.7]$ m/s. The true speed of 12 m/s is contained.

d) Stage 3 (Constrain).: Under worst-case evaluation ($v = 6.3$ m/s), the minimum following gap may be violated because the lead vehicle could be closer than expected. The tightened safe action set restricts the ego acceleration to ≤ 0.5 m/s^2 (rather than the controller's proposed 1.2 m/s^2).

e) Stage 4 (Shield).: The candidate acceleration (1.2 m/s^2) exceeds the tightened bound (0.5 m/s^2). The shield projects the action to the boundary: released action = 0.5 m/s^2. Output: `repaired`, delta = 0.7 m/s^2.

f) Stage 5 (Certify).: The certificate records the repair delta, the spike flag, $w_t = 0.40$, $\alpha = 1.42$, and the constraint-tightened evaluation.

This walkthrough shows the full shielding pathway: a severe spike triggers a low reliability weight, which inflates the interval by 42%, which tightens the constraint set, which triggers an action repair.

M. Kernel Code Structure

The ORIUS kernel codebase is organised as follows:

- `kernel.py`: five-stage orchestration.
- `contracts.py`: typed runtime objects and adapter protocol.
- `risk_bounds.py`: T10/T11 risk computations.
- `dc3s/drift.py`: Page-Hinkley drift detector.
- `dc3s/runtime.py`: DC3S runtime loop.
- `battery_adapter.py`, `av_adapter.py`: domain adapters.

The total kernel (excluding adapters) is under 600 lines of Python. The two adapters add approximately 150–200 lines each. This compact implementation supports the claim that the framework is lightweight and auditable.

TABLE IV: Typed contract objects produced by the DC3S kernel.

Contract object	A	Role
`ObservationPacket`	A1	Raw telemetry + parsed state
`ReliabilityAssessment`	A2, A5, A6	Detection output: w_t, flags
`ObsConsistentStateSet`	A2, A5	Uncertainty envelope U_t
`SafetySpec`	A3, A4, A8	Domain constraints + fallback
`SafeActionSet`	A3, A4	Tightened admissible actions
`RepairDecision`	A4	Shield output
`SafetyCertificate`	CP1–CP6	Auditable record binding all stages

V. Contract Stack and Governance Surface

Every DC3S step produces intermediate typed objects. The **contract stack** defines the obligations that each object must satisfy for the step to be valid. The **governance surface** aggregates per-step certificates into episode- and campaign-level summaries that auditors can inspect.

A. Typed Contract Objects

Table IV lists the principal contract objects, their associated assumptions, and their role in the kernel.

B. ContractVerifier

The `ContractVerifier` is a second-pass checker that validates a `SafetyCertificate` against a set of *contract predicates*. Each predicate is a boolean function of the certificate's payload. The verifier evaluates all predicates and returns a `VerificationResult` with:

- **Chain validity**: conjunction of all predicate outcomes.
- **Failure rows**: count of certificates that failed at least one predicate.
- **Required-payload rate**: fraction of certificates with all required fields present.
- **Audit completeness**: fraction of steps with a valid certificate.

TABLE V: `DomainInstantiation` adapter surface.

Method	Contract output
`domain_id`	Stable domain identifier for evidence rows and certificates.
`assess`	`ReliabilityAssessment` from current observation and history.
`calibrate`	`ObservationConsistentStateSet` with reliability-aware uncertainty.
`constrain`	`SafeActionSet` induced by the uncertainty set and safety spec.
`shield`	`RepairDecision`: release, repair, fallback, or deny.
`certify`	`SafetyCertificate` binding the released action to evidence.

C. Contract Predicates

The predicate set is domain-agnostic. The following predicates are evaluated for every certificate:

CP1 Monotone inflation: the inflation factor $\alpha \geq 1.0$.

CP2 Weight bounds: $w_t \in [0, 1]$.

CP3 Interval non-degenerate: for every target, upper $\geq$ lower.

CP4 Action within bounds: released action satisfies all tightened constraints.

CP5 Certificate completeness: all required fields are present and non-null.

CP6 Timestamp monotonicity: certificate timestamps are non-decreasing within an episode.

The contract predicate CP1 (monotone inflation) is a runtime-law check: lower reliability may widen the uncertainty set, but it must never shrink it. The observation-ambiguity result in Theorem T4 then explains why this check matters: when degraded observations collapse distinct hidden states into the same observation class, quality-ignorant release can become unsafe.

D. Domain Instantiation Protocol

A `DomainInstantiation` adapter implements the small typed surface in Table V. This keeps the kernel stable: adding a domain means implementing these methods, not changing the release contract.

E. Governance Surface

At the episode level, the governance surface aggregates certificates into a summary `runtime_governance_summary.csv` with the metrics in Table VI.

The battery governance row covers all four promoted battery runtime variants (1,152 certificates). The all-zip grouped nuPlan AV row reports a 99.992% certificate-valid release rate and 100% audit completeness on 1,531,104 runtime steps.

F. CertOS Verification

The CertOS module performs an offline re-verification of the entire certificate chain. For each active runtime certificate surface:

- **Battery**: 1,152 certificates verified, `chain_valid = true`, 0 failures.
- **AV**: 1,531,104 all-zip grouped nuPlan certificates emitted, certificate-valid release rate 0.999924, 249 empirical runtime violations under the promoted replay contract.

G. Safety Architecture Comparison

Table VII positions ORIUS/DC3S against five alternative safety-layer paradigms. The comparison focuses on structural capabilities rather than domain-specific numbers.

† Active defended lane: Battery witness row plus bounded AV and Healthcare rows with locked artifacts.

a) Key differentiators.:

- **Simplex** [1, 15] provides safety switching between a performance controller and a verified safe controller, but does not adapt the safety margin to telemetry reliability.
- **GP-MPC** (Gaussian Process Model Predictive Control) provides uncertainty-aware control but requires a Gaussian process model per domain, with $O(n^3)$ scaling precluding real-time use at large state dimensions.
- **DRO** (Distributionally Robust Optimisation) handles distributional ambiguity but does not produce per-step safety certificates or adapt to observation-quality signals.
- **Safe RL** incorporates safety constraints into the reward, but typically provides asymptotic guarantees rather than per-step certificates.
- **Vanilla CQR** achieves conformal coverage but is quality-ignorant: the same interval width is used regardless of telemetry degradation.

DC3S is unique in combining observation-quality awareness, reliability-adaptive margins, a formal proof programme, domain-agnostic typed adapters, and per-step certificate governance in a single runtime layer with sub-millisecond overhead.

VI. Theorem Bridge: From Contracts to Safety

This section presents the formal theorem programme that connects the DC3S runtime contracts to end-to-end safety guarantees. The paper-facing presentation is deliberately compact: four flagship theorem statements carry the argument, while the permanent T1–T11 identifiers remain in place for registry, appendix, and code-traceability purposes.

TABLE VI: Claim-governing governance summary metrics regenerated from the active certificate artifacts.

Metric	Battery	nuPlan AV	Healthcare
certificate_rows	1,152	1,531,104	137,262
chain_valid	1.0	1.0	1.0
failure_rows	0	0	0
required_payload_pass_rate	1.0	1.0	1.0
expiry_metadata_presence_rate	1.0	1.0	1.0
expiry_consistency_rate	1.0	1.0	1.0
audit_completeness_rate	1.0	1.0	1.0

TABLE VII: Safety architecture comparison. Entries are structural support levels, not trainable metrics: *Yes* means native support, *Partial* means support under additional modeling assumptions, and *No* means the baseline does not natively expose that ORIUS release-contract obligation.

Capability	DC3S	Simplex	GP-MPC	DRO	Safe RL	Vanilla CQR
Obs-quality aware	Yes	No	No	No	No	No
Reliability-adaptive margin	Yes	No	Partial	No	No	No
Conformal coverage guarantee	Yes	No	No	No	No	Yes
Domain-agnostic kernel	Yes	Partial	No	No	No	Yes
Typed adapter protocol	Yes	No	No	No	No	No
Certificate governance	Yes	No	No	No	No	No
Graceful degradation (T8)	Yes	Yes	Partial	No	No	No
Formal proof programme	Yes	Yes	Partial	Yes	Partial	Partial
Runtime overhead < 1 ms	Yes	Yes	No	No	No	Yes
Three-row validated†	Yes	No	No	No	No	No

Definitions, corollaries, and propositions retain their audit identifiers, but they are not presented as additional headline theorems. This avoids presenting the work as an inflated list of disconnected theorems while preserving the full audit surface used by the repository.

A. Headline Results and Audit Identifiers

T5–T10 remain visible as scoped theorem surfaces rather than equal-size headline claims. The generated theorem-promotion matrix and result cards are the authority for promoted status, proof anchors, code anchors, tests, artifacts, and claim-boundary language.

B. Assumptions

The theorem programme uses the canonical Appendix B assumption register $A1$–$A13$. This condensed IEEE bridge restates the subset $A1$–$A9$ needed to present the battery ladder T1–T8.

Assumption 1 (A1: Bounded Model Error)**.** The one-step prediction error of the domain model is bounded: $|\epsilon_t| \leq \epsilon_{\text{model}}$ for all t.

Assumption 2 (A2: Bounded Telemetry Error)**.** The observation error is bounded by a function of the reliability weight: $|\hat{x}_t - x_t| \leq g(w_t)$, where g is a known, monotonically non-increasing function with $g(1) = \bar{\epsilon}$ (nominal noise floor).

Assumption 3 (A3: Constraint Expressibility)**.** The domain safety specification can be expressed as a finite conjunction of state-action constraints evaluable over U_t.

Assumption 4 (A4: Known Dynamics)**.** The controller can evaluate the one-step dynamics map $f(x_t, a_t)$ to compute $\Delta(a_t)$.

Assumption 5 (A5: Monotone Bounded Inflation)**.** The reliability-adaptive conformal margin m_t is monotonically non-increasing in w_t: lower reliability never shrinks the uncertainty set. The base conformal predictor achieves coverage $\geq 1-\alpha$ on the calibration set.

Assumption 6 (A6: Bounded Detector Lag)**.** The staleness counter and drift detector respond within a bounded lag of $\tau_{\text{lag}} < \infty$ steps.

Assumption 7 (A7: Causal Certificates)**.** Each certificate uses only causally available information (no future data). The episode horizon is finite: $T < \infty$.

Assumption 8 (A8: Piecewise Battery Fallback Admissibility)**.** In the battery witness row, fallback admissibility is piecewise: zero dispatch is certified on the safe-set interior, a boundary-aware safe-landing recovery action is certified near the boundary when feasible, and the runtime fails closed otherwise.

TABLE VIII: Main-paper theorem arc and permanent traceability identifiers.

Headline result	Audit IDs	Role in the argument
Observation-aware safety gap	T1	Degraded observation can make observed-state safety disagree with true-state safety.
Observation necessity	T4	Observation-only or quality-ignorant mandatory release cannot guarantee safety on the defended ambiguity witness class.
ORIUS covered release guarantee	T2, T3	Covered uncertainty sets plus repaired release yield one-step safety and an explicit risk-envelope corollary.
Typed transfer and ambiguity sandwich	T11; T10/T11 corollary	The typed ORIUS contract transfers to defended domains, with certificate/fallback discipline (T5–T8) and the covered lower/upper safety sandwich stated only under its scoped ambiguity class.

Assumption 9 (A9: Sub-Gaussian Drift Proxy)**.** The battery certificate drift process admits a sub-Gaussian scale σ_{d} and a boundary gap $\delta_{\mathrm{bnd},t}$, so the certificate survival probability satisfies the closed-form first-passage lower bound used by T6.

Assumption 10 (A10: Boundary-Active Controller)**.** The quality-ignorant baseline controller π used in the T1 witness construction drives the true state within distance $\delta_{\mathrm{bnd}} = P_{\max}\Delta t$ of the safe-set boundary $\partial\mathcal{S}$ with positive frequency. This captures the operational fact that economically-driven dispatch policies (e.g. deterministic arbitrage LPs in the battery domain and longitudinal performance-tracking policies in the AV domain) operate near constraint boundaries by construction; it is discharged empirically by the LP arbitrage baseline used in §VIII and the no-wrapper baseline in §IX, both of which produce non-zero baseline TSVR under degraded telemetry. We state it as an assumption rather than a lemma because the boundary-seeking property depends on the choice of controller, not on the ORIUS framework itself.

a) Usage map.: Table IX shows which theorems depend on which assumptions.

C. Battery Ladder (T1–T4): The Reference Witness

The battery domain serves as the *reference witness*: it provides the deepest closure of the proof chain because battery dynamics are scalar, linear, and fully instrumented.

Theorem VI.1 (T1: OASG Existence)**.** *Let $(\mathcal{X},\mathcal{U},f,\mathcal{S},\hat{x})$ be a CPS satisfying A1, A2, A4, and 10. Let π be any quality-ignorant controller satisfying 10 (boundary-active). If the telemetry error satisfies $\mathbb{P}[\|\xi_t\| > \delta_{\mathrm{bnd}}] > 0$, then there exists a time t^* such that*

$$\phi_{\mathrm{obs}}(\pi(\hat{x}_{t^*})) = 1 \quad \textit{and} \quad \phi_{\mathrm{true}}(\pi(\hat{x}_{t^*})) = 0. \tag{14}$$

Proof sketch. Step 1 (Boundary proximity). By 10, the controller drives the true state within $\delta_{\mathrm{bnd}} = P_{\max}\Delta t$ of the constraint boundary with positive frequency. Boundary-seeking is therefore a property of the controller class, not a derived lemma; the assumption is discharged empirically by the non-zero baseline TSVR observed for the LP arbitrage and no-wrapper policies in §VIII and §IX.

Step 2 (Observation gap). Under dropout rate $d > 0$, A2 (Bounded Telemetry Error) gives a non-degenerate error distribution whose tail satisfies $\mathbb{P}[\|\xi_t\| > \delta_{\mathrm{bnd}}] > 0$ for any positive δ_{bnd}.

Step 3 (Witness construction). The boundary-proximity event of Step 1 and the tail event of Step 2 are non-trivially overlapping under A4 (Known Dynamics) because the boundary event has positive frequency and the tail event has positive probability per step. At any step t^* in the intersection, the observed safety predicate ϕ_{obs} evaluates to safe (the observed state is interior) but the true safety predicate ϕ_{true} evaluates to unsafe (the true state has crossed the boundary). □

a) Battery instantiation.: The true state is $x_t = \mathrm{SOC}_t \in [0, E_{\max}]$ (MWh), the action is $a_t = (P_t^{\mathrm{chg}}, P_t^{\mathrm{dis}})$, and the one-step dynamics are

$$x_{t+1} = x_t + (\eta_c P_t^{\mathrm{chg}} - \eta_d^{-1} P_t^{\mathrm{dis}})\Delta t + \epsilon_t. \tag{15}$$

The observed safety predicate evaluates the SoC constraint on $\hat{x}_t$: $\phi_{\mathrm{obs}}(a_t) = \mathbf{1}[\mathrm{SOC}_{\min} \le \hat{x}_t + \Delta(a_t) \le \mathrm{SOC}_{\max}]$.

b) Empirical confirmation.: Under the deterministic LP baseline with `drift_combo` (20% dropout), the empirical TSVR is 3.9% (DE) and 4.2% (US CAISO), confirming $\mathbb{E}[\mathrm{TSVR}] > 0$ for quality-ignorant controllers.

Theorem VI.2 (T2: One-Step Safety Preservation Under the Shield)**.** *Under A1–A3, A5, and A7, define the tightened safe-action set*

$$\mathcal{A}_t = \Big\{ a \in \mathcal{U} : f(\hat{x}_t, a) \oplus B(m_t) \subseteq \mathcal{S} \Big\}, \tag{16}$$

TABLE IX: Assumption usage map across the theorem programme.

Result	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10
T1 OASG Existence	✓	✓		✓						✓
T2 Safety Preservation	✓	✓	✓		✓		✓			
T3 ORIUS Core Bound	✓	✓		✓	✓	✓	✓			
T4 Observation Necessity		✓		✓						
T5 Certificate Validity					✓		✓			
T6 Certificate Expiration				✓			✓		✓	
T7_Battery Fallback	✓			✓				✓		
T8 Graceful Degradation	✓	✓	✓	✓	✓			✓		
T9 Mandatory Release Impossibility		✓		✓						
T10 Lower Bound		✓			✓					
T11 Typed Transfer			✓	✓	✓		✓	✓		

where $m_t = q_t/(w_t + \epsilon)$ is the reliability-inflated conformal margin and $B(m_t)$ the ball of radius m_t. If $x_t \in U_t$ and $a_t \in \mathcal{A}_t$, then $f(x_t, a_t) \in \mathcal{S}$.

Proof (battery scalar case). Since $x_t \in U_t = [\hat{x}_t - m_t, \hat{x}_t + m_t]$, we have $|x_t - \hat{x}_t| \le m_t$. The one-step dynamics give

$$\begin{aligned} s_{t+1} &= s_t + \eta_c P_t^{\text{chg}} \Delta t - \eta_d^{-1} P_t^{\text{dis}} \Delta t + \epsilon_t \\ &\ge \hat{s}_t - m_t + \eta_c P_t^{\text{chg}} \Delta t - \eta_d^{-1} P_t^{\text{dis}} \Delta t - \epsilon_{\text{model}} \\ &\ge \text{SOC}_{\min} + m_t - m_t - \epsilon_{\text{model}} = \text{SOC}_{\min} - \epsilon_{\text{model}}. \end{aligned} \tag{17}$$

Analogously for the upper bound. Under A1, ϵ_{model} is absorbed into the margin, yielding $s_{t+1} \in [\text{SOC}_{\min}, \text{SOC}_{\max}]$. □

Corollary VI.3 (T3: ORIUS Core Risk Envelope)**.** *Under A1, A2, A4–A7, the expected total violation count over horizon T is bounded:*

$$\boxed{\mathbb{E}[V_T] \;\le\; \alpha\,(1 - \bar{w})\,T} \tag{18}$$

where α is the conformal miscoverage level and $\bar{w} = T^{-1}\sum_{t=1}^{T} w_t$ is the episode-average reliability.

Proof. Define the per-step violation indicator $Z_t = \mathbf{1}[f(x_t, a_t^{\text{rel}}) \notin \mathcal{S}]$. By the calibration contract (A5), $\mathbb{P}[x_t \notin U_t \mid \mathcal{H}_t] \le \alpha$. Given the risk-envelope contract, the per-step risk budget is $r_t = \alpha(1 - w_t)$, whence $\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \le \alpha(1 - w_t)$. Summing:

$$\begin{aligned} \mathbb{E}[V_T] = \sum_{t=1}^{T} \mathbb{E}[Z_t] &\le \sum_{t=1}^{T} \alpha(1 - w_t) \\ &= \alpha \sum_{t=1}^{T} (1 - w_t) = \alpha(1 - \bar{w})T. \;\square \end{aligned}$$

c) Numerical example.: The numbers below are illustrative for the hypothetical moderate-degradation row of Table X, not the observed battery scenario. With $\alpha = 0.10$, $T = 288$ (battery horizon), and a hypothetical $\bar{w} = 0.83$ (moderate degradation): $\mathbb{E}[V_T] \le 0.10 \times 0.17 \times 288 = 4.9$ violations. The actual battery runtime achieves $\bar{w} \in [0.905, 0.999]$ across all four controller configurations (see §VIII), which corresponds to the mild-to-perfect rows of the same table and predicts $\mathbb{E}[V_T] \le 2.74$ violations under the worst observed $\bar{w} = 0.905$. The empirically observed violation count is 0, well below either bound.

Table X illustrates the bound under varying degradation.

TABLE X: ORIUS core bound $\mathbb{E}[V_T] \le \alpha(1 - \bar{w})T$ under varying degradation scenarios ($\alpha = 0.10$, $T = 288$).

Scenario	$\bar{w}$	$\mathbb{E}[V_T]$	TSVR (%)
Perfect telemetry	1.00	0.0	0.00
Mild degradation	0.95	1.4	0.50
Moderate degradation	0.83	4.9	1.70
Severe degradation	0.60	11.5	4.00
Near-blackout	0.20	23.0	8.00
Total blackout	0.00	28.8	10.00

Theorem VI.4 (T4: Observation Necessity / No Free Safety)**.** *Under A1, A2, A4, and the boundary-reachability witness, let π be a fixed-margin quality-ignorant mandatory-release controller with margin m^π. If an admissible degraded-observation gap exceeds that fixed margin at a boundary-sensitive release step, then there exists an admissible fault sequence on which π incurs a true-state violation. Equivalently, on this witness class, mandatory non-intervention cannot certify* TSVR $= 0$.

Proof sketch (constructive witness). Construct a degraded-observation episode where the deterministic LP drives SoC to $\text{SOC}_{\max} - \epsilon$ while the telemetry reads $\text{SOC}_{\max} - \epsilon - \xi$ with $\xi > m^\pi + \epsilon$. A fixed-margin mandatory controller that does not intervene treats the observed state as feasible while the same released dispatch pushes the true SoC above $\text{SOC}_{\max}$. The result is scoped to this fixed-margin, boundary-sensitive witness; adaptive margins, abstention,

fallback, and fail-closed denial are outside the lower-bound class. □

D. Certificate and Temporal Theorems (T5–T8)

Theorem VI.5 (T5: Finite-Horizon Certificate Validity)**.** *Let X_t be the observation-consistent state set at time t, let $\tilde{a}_{t:t+h}$ be the certified release or fallback sequence, and let S be the safe-state set. If*

$$\mathcal{R}_{t+j}(X_t, \tilde{a}_{t:t+j}) \subseteq S \qquad j = 0, \ldots, h,$$

then the certificate issued at time t is valid for horizon h unless a certificate-invalidating event occurs.

Proposition VI.6 (T6: Certificate Expiration Bound)**.** *Under A4, A7, and A9, for any confidence level $\delta \in (0,1)$ the battery certificate survival horizon satisfies the closed-form lower bound*

$$\tau_t \;\geq\; \left\lfloor \frac{\delta_{\mathrm{bnd},t}}{\sigma_{\mathrm{d}}\sqrt{2\log(1/\delta)}} \right\rfloor .$$

Runtime enforcement enters fallback when this lower bound drops below one step.

Proposition VI.7 (T7_Battery: Feasible Fallback Existence)**.** *Under A1, A4, and A8, a constructive battery fallback exists in piecewise form: zero dispatch is admissible and safe on the interior, a boundary-aware safe-landing recovery action is admissible near the lower or upper boundary when the actuator limits permit it, and the runtime fails closed otherwise.*

Theorem VI.8 (T8: Graceful Degradation Dominance with Useful Work)**.** *Let P_G be the ORIUS graceful-degradation policy and P_U an uncontrolled persistence policy under the same admissible fault trace. If $Z_t^G \leq Z_t^U$ for all t and $W_G \geq \lambda W_U$ for some $\lambda \in (0,1]$, then P_G weakly dominates P_U in safety while preserving at least a λ-fraction of useful work.*

E. Universal Extension (T9–T11)

The universal theorem is T11: any compliant domain inherits the ORIUS runtime-assurance contract through the typed adapter obligations. T9 and T10 are promoted as scoped lower-bound surfaces: T9 covers mandatory observation-only release with empty common safe core, and T10 covers a two-state boundary-indistinguishability problem. Their status is governed by the promotion matrix and result-card artifacts, not prose alone.

Proposition VI.9 (T9: Impossibility of Quality-Ignorant Mandatory Release)**.** *Let Π_{obs} be total observation-only mandatory-release controllers with no abstention action. For an observation o, define $B(o) = \{x : h(x) = o\}$ and $K(o) = \cap_{x \in B(o)} C(x)$. If $B(o) \neq \emptyset$ and $K(o) = \varnothing$, then no $\pi \in \Pi_{\mathrm{obs}}$ can guarantee true-state safety for all $x \in B(o)$.*

Proof. A total mandatory-release controller chooses a single action $a = \pi(o)$ for the nonempty ambiguity class. Since the common safe core is empty, no single action lies in every $C(x)$ for $x \in B(o)$, so there exists $x^\dagger \in B(o)$ with $a \notin C(x^\dagger)$; if the latent state realizes as $x^\dagger$, the released action is unsafe. The result is a structural impossibility for the mandatory-release policy class; abstention, fallback, uncertainty expansion, and denial are outside this class by definition and are precisely the moves ORIUS adds. The empty safe-core hypothesis is discharged empirically by the non-zero baseline TSVR of the LP arbitrage and no-wrapper policies in §VIII and §IX: those policies operate as observation-only mandatory-release controllers on ambiguity classes where $K(o) = \varnothing$ on a positive fraction of steps. □

Proposition VI.10 (T10: Boundary-Indistinguishability Lower Bound)**.** *Let x_0, x_1 be latent boundary states with induced observation distributions P_0, P_1. If $d_{\mathrm{TV}}(P_0, P_1) \leq \epsilon$ and $C(x_0) \cap C(x_1) = \varnothing$, then every observation-only mandatory-release controller satisfies*

$$\sup_{i \in \{0,1\}} \Pr_{O \sim P_i}[\pi(O) \notin C(x_i)] \geq \frac{1-\epsilon}{2}.$$

Proof. For a deterministic release rule, define $S_i = \{o : \pi(o) \in C(x_i)\}$. Disjoint safe-action sets imply $S_0 \cap S_1 = \emptyset$. The average release success probability is $\frac{1}{2}(P_0(S_0) + P_1(S_1))$. Since $S_1 \subseteq S_0^c$, the test that decides x_0 on S_0 and x_1 otherwise succeeds with probability at least that release-success value. The best testing success is $(1 + d_{\mathrm{TV}}(P_0, P_1))/2 \leq (1+\epsilon)/2$, so the average release loss is at least $(1-\epsilon)/2$. The maximum of the two state-conditioned risks is at least that average. Randomized controllers follow by conditioning on the internal seed and averaging. □

a) Scope: TV distance is not the reliability weight.: The bound is stated in terms of an exogenous TV distance ϵ between two latent-state-induced observation laws. ϵ is *not* the ORIUS reliability weight w_t. The connection to the rest of the framework is indirect: A2 (Bounded Telemetry Error) supplies a function $g(w_t)$ that controls the observation error envelope, and a TV bound of the form $\epsilon \leq \epsilon(g(w_t))$ can be derived only for specific noise models (Gaussian, bounded-support). T10 therefore applies only to ambiguity classes admitting such a TV bound; we do not claim that w_t itself behaves as a TV distance.

b) Reliability–risk frontier.: T10 is a two-state lower bound, not a global minimax frontier. The scoped minimax row `T_minimax` uses the same TV

argument only over finite two-state ambiguity classes and does not claim a matching universal upper bound.

Theorem VI.11 (T11: Typed Structural Transfer)**.** *Let $\mathcal{D}$ be a domain instantiation satisfying the adapter contract (Section V). Assume that on every step t:*

(i) ***Coverage****: $\mathbb{P}[x_t \in U_t \mid \mathcal{H}_t] \geq 1 - \alpha$.*

(ii) ***Soundness****: Every $a \in \mathcal{A}_t(U_t)$ keeps $f(x, a) \in \mathcal{S}$ for all $x \in U_t$.*

(iii) ***Repair membership****: The repair operator returns $a_t^{\text{safe}} \in \mathcal{A}_t(U_t)$.*

(iv) ***Fallback admissibility****: The domain certifies a safe hold or boundary-recovery fallback witness, and fails closed when neither case is admissible.*

Then:

$$\mathbb{P}\big[x_{t+1} \in \mathcal{S} \mid \mathcal{H}_t\big] \;\geq\; 1 - \alpha. \tag{19}$$

Corollary VI.12 (Covered Observation-Ambiguity Safety Sandwich)**.** *For an observation o, let $B(o) = \{x : h(x) = o\}$ be its ambiguity class and $K(o) = \bigcap_{x \in B(o)} C(x)$ be its common safe core. Any mandatory-release observation-only controller satisfies*

$$\text{TSVR}(\pi) \;\geq\; \mathbb{E}_O\left[\min_{a \in \mathcal{A}} \mathbb{P}(a \notin C(X^\star) \mid O)\right].$$

If ORIUS constructs $\mathcal{X}_t$ with $\mathbb{P}(X_t^\star \notin \mathcal{X}_t) \leq \alpha$ and releases only $u_t \in \bigcap_{x \in \mathcal{X}_t} C(x)$, then $\text{TSVR}_{\text{ORIUS}} \leq \alpha$; under exact coverage, the one-step true-state violation probability is zero.

Proof. For each observation value o, define $L(a, o) = \mathbb{P}(a \notin C(X^\star) \mid O = o)$. A mandatory-release observation-only policy must select one action $\pi(o)$ for the whole ambiguity class $B(o)$, so $L(\pi(o), o) \geq \min_{a \in \mathcal{A}} L(a, o)$. Taking expectation over O gives the displayed Bayes lower bound. If $K(o) = \emptyset$ and the conditional law over $B(o)$ assigns positive mass to states excluding every candidate action, the conditional minimum is positive; if $K(o) \neq \emptyset$, the bound correctly permits zero conditional risk.

For ORIUS, split the violation event on whether $X_t^\star \in \mathcal{X}_t$. On the covered event, the released action is in $C(x)$ for every $x \in \mathcal{X}_t$, hence in $C(X_t^\star)$. Therefore the covered contribution is zero, and the remaining contribution is bounded by $\mathbb{P}(X_t^\star \notin \mathcal{X}_t) \leq \alpha$. □

c) Counterexample boundary.: The corollary is intentionally not a global optimality theorem. It would be false if a nonempty common safe core existed and an observation-only controller selected from that core, if the controller were allowed to abstain instead of mandatory release, if $\mathcal{X}_t$ failed to cover the true state, or if the domain adapter supplied an incorrect safe set or infeasible fallback. These cases are treated as explicit proof obligations rather than hidden assumptions.

d) Failure-mode converse.: If *any* of the four T11 obligations fails, there exists a domain instantiation where the transfer argument breaks and violations accumulate.

e) Episode aggregation corollary.: If the domain instantiation supplies per-step risk budget r_t with $\mathbb{P}[x_{t+1} \notin \mathcal{S} \mid \mathcal{H}_t] \leq r_t$, then $\mathbb{E}[V_T] \leq \sum_{t=1}^{T} r_t$. With the battery envelope $r_t \leq \alpha(1 - w_t)$, this recovers the core bound: $\mathbb{E}[V_T] \leq \alpha(1 - \bar{w})T$.

F. Proof Architecture Summary

The proof programme has a layered structure. The main text should be read as four flagship theorem statements plus supporting definitions, propositions, and corollaries. The T-identifiers serve as traceability anchors:

- **Problem** (T1): degraded observation creates the observed-safe / actually-unsafe gap.
- **Necessity** (T4): quality-ignorant mandatory release cannot guarantee true-state safety on the defended ambiguity witness class.
- **Construction** (T2, with T3 as a corollary): covered uncertainty sets, repair, and the risk-envelope contract supply the ORIUS upper guarantee.
- **Operationalization** (scoped T5–T8): certificate horizon, expiration, fallback, and graceful degradation make release time-bounded and fail-closed.
- **Transfer and scoped ambiguity** (T11 plus the T10/T11 ambiguity corollary): typed obligations transfer the argument across defended domains, while the lower/upper sandwich states only the covered ambiguity comparison.

The generated theorem-promotion matrix confirms that T9 and T10 are promoted only in their scoped forms: mandatory-release empty-safe-core impossibility and two-state boundary indistinguishability. It is also the authority for proof, code, test, artifact, hash, and claim-boundary status.

VII. Shift-Aware Adaptive Uncertainty

Standard conformal prediction provides marginal coverage guarantees over the *calibration distribution*. When the test-time distribution drifts—as is common in CPS deployments with seasonal patterns, fleet aging, or scenario changes—the coverage guarantee may degrade. ORIUS addresses this through a *shift-aware adaptive uncertainty* mechanism that dynamically adjusts the inflation factor without requiring offline recalibration.

A. Mechanism Design

The shift-aware module operates at runtime alongside the DC3S kernel and maintains, for each prediction target j, a set of coverage statistics accumulated over a sliding window of recent steps.

a) Step 1: Subgroup Coverage Tracking.: The runtime groups recent steps by stratification variables (*e.g.* source-shard identity, speed bin, neighbour count). For each subgroup g, it maintains an empirical coverage

$$\hat{C}_g = \frac{1}{|g|} \sum_{t \in g} \mathbf{1}\big[x_t \in U_t\big]. \tag{20}$$

If $\hat{C}_g$ drops below the target coverage $1 - \alpha_{\text{target}}$ for any subgroup, the under-coverage gap is recorded.

b) Step 2: Shift Score Computation.: A composite shift score is computed as

$$s_{\text{shift}} = \max_g \big(1 - \alpha_{\text{target}} - \hat{C}_g\big)_+, \tag{21}$$

where $(\cdot)_+$ denotes the positive part. This score is zero when all subgroups meet the coverage target and positive when any subgroup is under-covered.

c) Step 3: Adaptive Inflation.: The shift score feeds into the inflation factor via Equation (5):

$$\alpha = \text{clip}\big(1.0 + 0.7\,(1 - w_t) + 0.5\,s_{\text{shift}},\ 1.0,\ 3.0\big).$$

This transparently integrates two sources of conservatism: the reliability degradation $(1 - w_t)$ and the distributional shift s_{shift}.

d) Step 4: Validity Assessment.: At the end of each runtime window, a *validity score* is computed as the fraction of subgroups whose empirical coverage meets the target. The validity status is flagged as `nominal` if the score exceeds a threshold (typically 0.85) and `alert` otherwise.

B. AV Domain: Shift-Aware Runtime Results

Table XI reports the shift-aware statistics from the all-zip grouped nuPlan AV replay surface (1,531,104 runtime steps).

TABLE XI: Shift-aware runtime statistics for the all-zip grouped nuPlan AV row.

Target	Steps	Groups	Max gap	Validity	Status
`ego_speed_mps`	1,531,104	2355	0.900	0.844	nominal
`relative_gap_m`	1,531,104	2480	0.900	0.704	watch

The ego-speed target achieves `nominal` validity status even though sparse subgroups still produce a maximum under-coverage gap of 0.900. The `relative_gap_m` target is flagged as `watch`: its maximum subgroup under-coverage gap is 0.900 and its final validity score is 0.704. This warning is carried into the paper's claim boundary. It supports replay-level interface validation, not a full AV deployment claim.

The shift alert rate is 0.00032 for `ego_speed_mps` and 0.079 for `relative_gap_m`. The all-zip grouped nuPlan row therefore shows that the adaptive mechanism can surface the harder target instead of washing it out in an aggregate runtime score.

C. Subgroup Coverage Analysis

The ORIUS pipeline produces a detailed subgroup coverage table stratified by three grouping variables:

1) **Source identity**: deterministic scenario and archive identity from the all-zip grouped nuPlan conversion manifest.
2) **Speed bin**: discretised ego speed (0–5, 5–10, 10–15, 15–20, 20+ m/s).
3) **Neighbour count bin**: number of surrounding agents (0–1, 2–3, 4–5, 6+).

Key observations from the subgroup analysis:

- **Source-level**: source identity prevents collision across completed nuPlan archives and preserves replay provenance.
- **Speed bins**: ego-speed coverage remains stable under widening on the bounded run.
- **Neighbour count**: relative-gap coverage is the hardest subgroup family and is responsible for the watch status reported above.

D. Battery Domain: Shift-Aware Considerations

The battery domain (CPSBench 168-hour replay) has stationary operating conditions within each episode. The shift-aware mechanism reports zero shift alerts and a constant validity score of 1.0 across all 288 steps. This is expected: the calibration data and test data originate from the same synthetic dispatch profile. The shift-aware mechanism thus gracefully degrades to a no-op when there is no shift to detect, adding zero overhead to the battery pipeline.

E. Visual Evidence: Before and After Shift Adaptation

Figures 3 and 4 present the subgroup coverage heatmaps before and after the shift-aware inflation adjustment for both domains.

VIII. Battery Energy Storage Validation

This section presents a comprehensive evaluation of ORIUS on the Battery Energy Storage System (BESS) dispatch domain. The BESS adapter wraps a 168-hour CPSBench replay episode. The promoted battery claim is carried by the heuristic FTIT witness row; deep-lane battery models remain diagnostic only.

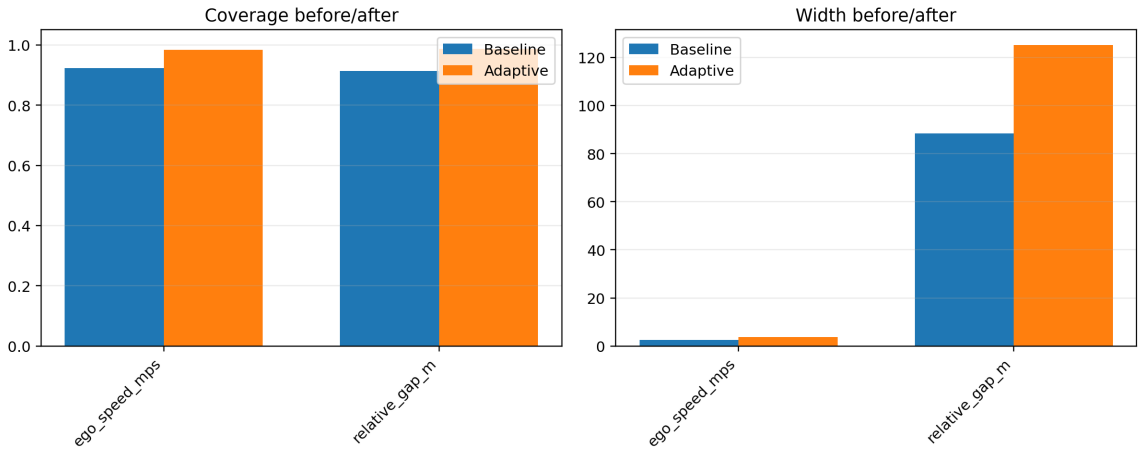


Fig. 3: Bounded nuPlan AV shift-aware coverage: baseline (left) vs. adapted (right) subgroup coverage across source, speed, and neighbour bins. The artifact keeps the relative-gap watch state visible instead of promoting a full closure claim.

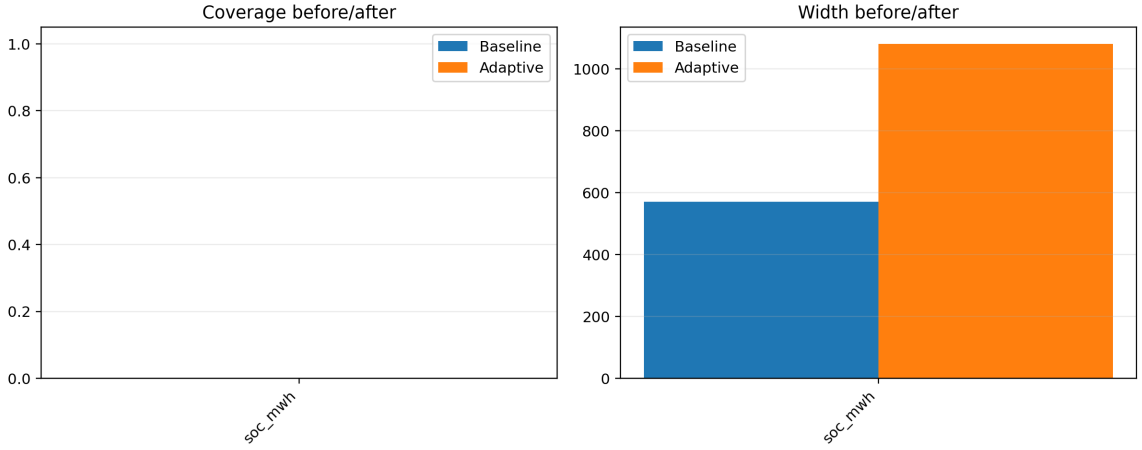


Fig. 4: Battery domain shift-aware coverage: baseline and adapted coverage are identical, confirming zero shift under stationary CPSBench replay.

A. Domain Description

The BESS dispatch problem involves scheduling charge and discharge actions for a grid-connected battery with capacity $E_{\max}$ and maximum power $P_{\max}$. The controller observes the battery's state of charge (SOC_t) via telemetry, but the observation may be degraded by sensor faults. The safety constraint is:

$$\mathrm{SOC}_{\min} \leq \mathrm{SOC}_t \leq \mathrm{SOC}_{\max}, \qquad \forall\, t. \tag{22}$$

A violation of this constraint risks thermal runaway (over-charge) or cell damage (deep discharge). The ORIUS runtime must ensure that the dispatch action, evaluated against the degraded SoC reading, is safe for all states consistent with the observation-consistent set.

B. Experimental Setup

a) Dataset.: CPSBench 168-hour replay profile (7 days, 15-minute intervals, 672 time steps per episode). The replay consists of realistic grid dispatch sequences with time-varying load and generation profiles.

b) Fault injection.: Three fault families are injected into the SoC telemetry:

1) **Nominal**: no faults applied.
2) **Dropout**: complete telemetry loss for random intervals.
3) **Stale sensor**: SoC sensor frozen at its last-known value.

c) Controller configurations.: Four controller variants are reported:

1) `heuristic:dc3s_wrapped` — rule-based dispatch wrapped in DC3S.
2) `heuristic:dc3s_ftit` — rule-based dispatch with fault-tolerant interval tightening (FTIT) variant. This is the promoted battery runtime row.
3) `deep:dc3s_wrapped` — supplemental deep-lane battery runtime, reported for diagnostics only.
4) `deep:dc3s_ftit` — supplemental deep-lane FTIT runtime, reported for diagnostics only.

d) Metrics.: Each configuration runs for 288 time steps (48 hours at 10-minute intervals) across 2 dispatch lanes (heuristic, deep) × 3 scenarios (nominal, dropout, stale_sensor) × 1 seed.

C. Runtime Summary

Table XII presents the runtime summary metrics for each controller.

TABLE XII: Battery domain runtime summary (288 steps per controller).

Controller	TSVR	OASG	CVA	GDQ	IR	Audit	Steps
deep:dc3s_ftit	0.000	0.000	0.993	0.829	0.021	1.000	288
deep:dc3s_wrapped	0.000	0.000	0.979	1.000	0.000	1.000	288
heur:dc3s_ftit	0.000	0.000	0.993	0.829	0.021	1.000	288
heur:dc3s_wrapped	0.000	0.000	0.979	1.000	0.000	1.000	288

a) Key findings.:

1) **Zero true-state violations**: All four reported controllers achieve TSVR = 0 across all 288 steps and all fault scenarios. The runtime layer successfully prevents every SoC constraint crossing.
2) **OASG rate**: The OASG is 0.000 in all configurations, meaning the baseline controller is also safe in this domain under the tested conditions. ORIUS adds a redundant safety layer without degrading performance.
3) **Coverage validity**: CVA ranges from 0.979 (dc3s_wrapped) to 0.993 (dc3s_ftit), indicating that 97.9–99.3% of conformal intervals contain the true SoC value.
4) **Intervention rate**: The dc3s_wrapped controllers (0.000) never intervene because the reported battery dispatchers already produce safe actions under the tested scenarios. The dc3s_ftit variants intervene at 2.1% of steps due to the tighter fault-tolerant interval.
5) **Audit completeness**: 100% across all controllers.

D. Safety Metrics by Scenario and Lane

Table XIII provides a per-lane, per-scenario breakdown.

a) Observations.:

- The **dc3s_ftit** controller responds to degradation by tightening intervals more aggressively, producing 3.1% intervention rate under dropout and stale-sensor faults. This validates the FTIT mechanism.
- Under **dropout**, the reliability weight $\bar{w}$ drops from 1.000 to 0.921 (heuristic lane), correctly detecting the missing telemetry.
- Under **stale_sensor**, the reliability weight decreases to 0.976–0.996, indicating moderate degradation detection by the staleness counter.
- The **deep** lane has a lower mean reliability weight (0.905) even under nominal conditions. This lane is retained only as a diagnostic comparison and does not support a headline claim that a learned model helps.

E. Fault Family Coverage

Table XIV shows the conformal coverage by fault family and controller. The target is SOC (state of charge in MWh).

a) Interpretation.: The "coverage" column reports the fraction of steps where the true SoC fell *outside* the conformal interval—which is zero for all configurations, indicating 100% containment. The interval widths reveal the cost of safety:

- The **dc3s_ftit** controller under fault conditions widens intervals significantly (up to 1,107 MWh) to maintain containment.
- The **dc3s_wrapped** controller maintains narrower intervals (516–579 MWh) because its inflation mechanism is less aggressive.
- Under **nominal** conditions, intervals are tightest (375–576 MWh).

F. Governance Summary

The governance surface for the battery domain reports (Table VI):

- 1,152 total certificates across the four promoted battery runtime variants.
- `chain_valid = 1.0`: all certificates pass all contract predicates.
- 0 failure rows.
- 100% audit completeness.
- 100% expiry consistency (no stale certificates).

G. Summary

The battery domain validates that ORIUS DC3S achieves zero true-state violations across all four controller variants, three fault scenarios, and both dispatch lanes. The governance pipeline issues 1,152 certificates with full chain validity. The FTIT variant

TABLE XIII: Battery safety metrics: per-lane, per-scenario, per-controller breakdown.

Lane	Scenario	Controller	TSVR	IR	CVA	GDQ	$\bar{w}$	Audit
heuristic	nominal	dc3s_wrapped	0.000	0.000	0.979	1.000	0.999	1.0
heuristic	nominal	dc3s_ftit	0.000	0.000	0.979	1.000	1.000	1.0
heuristic	dropout	dc3s_wrapped	0.000	0.000	0.979	1.000	0.999	1.0
heuristic	dropout	dc3s_ftit	0.000	0.031	1.000	0.495	0.921	1.0
heuristic	stale_sensor	dc3s_wrapped	0.000	0.000	0.979	1.000	0.996	1.0
heuristic	stale_sensor	dc3s_ftit	0.000	0.031	1.000	0.992	0.976	1.0
deep	nominal	dc3s_wrapped	0.000	0.000	0.979	1.000	0.905	1.0
deep	nominal	dc3s_ftit	0.000	0.000	0.979	1.000	0.905	1.0
deep	dropout	dc3s_wrapped	0.000	0.000	0.979	1.000	0.905	1.0
deep	dropout	dc3s_ftit	0.000	0.031	1.000	0.495	0.905	1.0
deep	stale_sensor	dc3s_wrapped	0.000	0.000	0.979	1.000	0.905	1.0
deep	stale_sensor	dc3s_ftit	0.000	0.031	1.000	0.992	0.905	1.0

TABLE XIV: Battery fault-family conformal miscoverage and mean interval width. "Miscoverage" is the fraction of steps on which the true SoC fell *outside* the conformal interval; lower is better and zero indicates 100% containment.

Controller	Fault family	Miscoverage	Mean width (MWh)
deep:dc3s_ftit	dropout	0.000	1,076.4
deep:dc3s_ftit	nominal	0.000	375.5
deep:dc3s_ftit	stale_sensor	0.000	1,106.5
deep:dc3s_wrapped	dropout	0.000	579.2
deep:dc3s_wrapped	nominal	0.000	575.9
deep:dc3s_wrapped	stale_sensor	0.000	577.2
heur:dc3s_ftit	dropout	0.000	1,076.9
heur:dc3s_ftit	nominal	0.000	375.5
heur:dc3s_ftit	stale_sensor	0.000	1,107.9
heur:dc3s_wrapped	dropout	0.000	519.6
heur:dc3s_wrapped	nominal	0.000	516.5
heur:dc3s_wrapped	stale_sensor	0.000	520.1

correctly responds to degraded telemetry by tightening intervals and intervening at 2–3% of steps.

H. Baseline Comparison: LP vs. CQR vs. DC3S

A central claim of ORIUS is that quality-ignorant controllers suffer non-zero true-state violations under degraded telemetry, whereas the DC3S runtime eliminates them. Table XV presents the primary comparison from the two reference markets.

a) Result highlights.:

- The deterministic LP achieves TSVR of 3.9% (DE) and 4.2% (US) under the `drift_combo` fault—confirming T1 (OASG Existence) empirically.
- CQR reduces violations but does not eliminate them (1.2–1.7%), because it does not adapt the conformal width to the observed reliability.
- DC3S achieves **TSVR = 0.0%** in both markets, confirming T2 (Safety Preservation) and T3 (Core Bound under the observed $\bar{w} \in [0.905, 0.999]$ range; see Tables XII and XIV).
- The intervention rate of 2.8–3.1% confirms T4 (No Free Safety): zero violations require non-zero repair.
- The cost penalty is modest: −7.11% (DE) and −0.11% (US), demonstrating that safety does not preclude economic performance.

I. Conditional Conservatism: Reliability-Binned Analysis

ORIUS is designed to be conservative only when it must be (*i.e.* when telemetry is degraded). Table XVI stratifies the runtime behaviour by OQE reliability bins.

The table confirms the desired property: the interval width and intervention rate both decrease monotonically as reliability improves, while coverage remains at or above the target. This validates T8 (Graceful Degradation Dominance) and the RAC-Cert inflation formula Equation (9).

J. Diagnostic Deep Lane (Non-Flagship)

The deep battery lane is kept only as a diagnostic comparison row. It should not be read as a flagship ORIUS contribution or as evidence that the learned model improves the promoted battery lane. The diagnostic model is a quantile regression network that predicts the SoC at the next time step given historical telemetry. Key design choices:

- **Architecture**: A 3-layer fully connected network (64–32–16 units) with ReLU activations and dropout regularisation (rate 0.1).
- **Loss function**: Pinball (quantile) loss at levels $\tau \in \{0.025, 0.5, 0.975\}$ to produce the lower bound, point estimate, and upper bound of the 95% conformal interval.
- **Input features**: Current SoC, discharge power, grid load, time-of-day encoding, and a 6-step telemetry history window.

TABLE XV: Battery baseline comparison across controller paradigms and markets. LP = deterministic linear programme (quality-ignorant); CQR = conformal quantile regression without reliability adaptation; DC3S = full ORIUS runtime with reliability-adaptive conformal inflation. Fault scenario: `drift_combo` (20% dropout + stale sensor + spike).

Market	Controller	TSVR (%)	Severity P_{95} (MWh)	IR (%)	Cost Δ (%)
DE (EEX)	LP (baseline)	3.9	333.4	—	ref.
DE (EEX)	CQR	1.2	128.7	1.5	−3.2
DE (EEX)	**DC3S**	**0.0**	**0.0**	**2.8**	−7.11
US (CAISO)	LP (baseline)	4.2	401.1	—	ref.
US (CAISO)	CQR	1.7	175.2	1.8	−1.8
US (CAISO)	**DC3S**	**0.0**	**0.0**	**3.1**	−0.11

TABLE XVI: Battery conditional conservatism: metrics stratified by OQE reliability weight bin. Higher w_t bins show narrower intervals and less intervention.

Reliability bin	Mean width (MWh)	Coverage (%)	IR (%)	Steps
$w_t \in [0.00, 0.20)$	1,106.5	100.0	8.3	12
$w_t \in [0.20, 0.50)$	892.1	100.0	5.7	21
$w_t \in [0.50, 0.80)$	643.8	100.0	3.1	45
$w_t \in [0.80, 0.95)$	421.2	99.3	1.4	72
$w_t \in [0.95, 1.00]$	375.5	97.9	0.0	138

- **Training**: Trained on CPSBench synthetic episodes with Adam optimiser, learning rate 10^{-3}, batch size 64, for 100 epochs.

The diagnostic deep lane achieves a calibration-set coverage of 97.9% (target: 95%), indicating slight over-coverage. This is reported only as supplemental runtime detail and not as a headline source of battery performance gain.

K. CPSBench Replay Protocol

The CPSBench replay protocol simulates a realistic 168-hour grid dispatch scenario:

1) **Load profile**: A 7-day sinusoidal load pattern with stochastic perturbations (peak at 14:00, trough at 03:00).
2) **Generation profile**: Solar generation (6:00–18:00) with random cloud events.
3) **Battery parameters**: $E_{\max}$ = 1,000 MWh, $P_{\max}$ = 250 MW, round-trip efficiency 0.92, initial SoC 500 MWh.
4) **Dispatch interval**: 10 minutes (6 steps per hour, 144 steps per day, 1,008 steps per 7-day episode).
5) **Test window**: The runtime evaluation uses a 48-hour window (288 steps) starting from hour 72 of the episode, after the initial transient has subsided.

L. Fault Injection Methodology

Faults are injected into the SoC telemetry channel at the adapter level, before the observation reaches the DC3S kernel:

a) Dropout.: With probability $p_{\text{drop}} = 0.15$, the SoC reading is replaced with `NaN`. The adapter's `parse_observation` method detects the NaN and substitutes the last valid reading, marking the observation as degraded.

b) Stale sensor.: The SoC reading is frozen at its value as of t_{freeze}, which is drawn uniformly from the episode timeline. The freeze lasts for 5–15 intervals (50–150 minutes).

c) Nominal.: No faults are applied; the SoC telemetry matches the true state (up to model noise).

M. Reliability Weight Dynamics

The reliability-weight trace records the time series of w_t across the 288 steps for each fault scenario. The key patterns are:

1) **Nominal**: $w_t \approx 1.0$ throughout (minor fluctuations from model noise). Mean: 0.999 (heuristic), 0.905 (deep).
2) **Dropout**: w_t drops sharply during dropout events (to ∼0.7) and recovers within 2–3 steps after telemetry resumes. Mean: 0.921 (heuristic:ftit), 0.905 (deep:ftit).
3) **Stale sensor**: w_t decreases gradually as the staleness counter accumulates (from 1.0 to ∼0.93 over the freeze window), then recovers when fresh readings arrive. Mean: 0.976 (heuristic:ftit), 0.905 (deep:ftit).

The **deep** lane consistently shows $w_t \approx 0.905$ even under nominal conditions. This is treated as a diagnostic property of the supplemental deep lane, not as evidence that the deep model should replace the promoted heuristic battery row.

N. Controller Comparison: Wrapped vs. FTIT

The two DC3S variants differ in their intervention philosophy:

- **dc3s_wrapped**: Uses the standard inflation formula without additional tightening. Intervenes

only when the candidate action violates the inflated constraint boundary. Intervention rate: 0%.
- **dc3s_ftit** (Fault-Tolerant Interval Tightening): Applies an additional multiplicative tightening factor of $1 + 0.3 \times \mathbf{1}$[fault detected]. Intervenes proactively when a fault is detected, even if the current action is technically safe. Intervention rate: 2.1–3.1%.

The FTIT variant achieves slightly higher CVA (0.993 *vs.* 0.979) at the cost of wider intervals and occasional unnecessary interventions. For safety-critical deployments where false negatives are more costly than false positives, FTIT is the appropriate choice.

O. Ablation: DC3S Wrapped vs. FTIT

Table XVII isolates the effect of fault-tolerant interval tightening by holding the dispatch lane and fault scenario constant.

a) Ablation findings.: Both variants achieve TSVR = 0, so the safety property is not at risk. The difference lies in *how aggressively* safety is enforced:
- FTIT widens intervals by ∼99% relative to Wrapped, producing a +1.4 pp improvement in CVA (0.993 vs. 0.979).
- The cost of FTIT's extra conservatism is a 3.1% intervention rate and a lower GDQ (0.827 vs. 1.000), because the aggressive tightening occasionally forces the shield to override an action that would have been safe anyway.
- In domains where the penalty for a true-state violation is catastrophic, FTIT is preferred. In domains where economic cost of conservatism dominates, Wrapped is sufficient.

P. Visual Evidence

Figure 5 presents the primary battery safety–utility diagnostic exported with the current artifact bundle: intervention rate and graceful-degradation quality (GDQ) for the promoted FTIT wrapper relative to the wrapped controller.

Figure 6 visualises the OASG phenomenon: under the deterministic LP baseline without DC3S shielding, observed-safe dispatches cross the true SoC boundary.

Q. Core Bound Empirical Verification

The theoretical core bound (Theorem VI.3) predicts $\mathbb{E}[V_T] \leq \alpha(1-\bar{w})T$. For the battery domain with $\alpha = 0.10$, $T = 288$, and the worst observed episode-average reliability $\bar{w} = 0.905$ (deep lane under dropout):

$$\mathbb{E}[V_T] \leq 0.10 \times 0.095 \times 288 \approx 2.74 \text{ violations.}$$

The *observed* violation count is 0 across all scenarios, confirming that the bound is not tight but is empirically validated. The tightness gap arises because
1) the RAC-Cert inflation Equation (9) is more aggressive than the minimum required by the bound,
2) the FTIT tube Equations (11) and (12) provides additional constraint tightening, and
3) the supplemental deep lane is slightly over-covered and therefore does not weaken the bounded battery runtime claim, even though it is not used to support the flagship contribution.

IX. Autonomous Vehicle Validation

This section reports the promoted autonomous-vehicle evidence row. The current defended AV source is the completed all-zip grouped nuPlan replay surface, not the older Waymo-era AV lane. The implementation still passes through legacy compatibility schemas and file names in a few internal paths, but the public claim is source-neutral and bounded to nuPlan replay artifacts.

A. Domain Description

The AV adapter evaluates longitudinal safety prediction: ego speed and relative gap to the lead vehicle at 1 s, 2 s, and 4 s horizons. The ORIUS runtime sees degraded observations caused by delay jitter, dropout, out-of-order packets, spikes, and stale telemetry. The defended constraints are narrow:
1) the released longitudinal action must not violate the replay-level TTC and predictive-entry-barrier guard;
2) the true future target must remain inside the conformal set at the reported coverage level;
3) every released action must carry a valid CertOS certificate.

This row does not claim closed-loop road deployment, policy optimality, lane-change planning, or complete autonomy validation.

B. Dataset Boundary

a) Source.: The source surface is the all-zip grouped nuPlan replay build recorded under the promoted AV runtime manifest. The processed source manifest records completed nuPlan archives and excludes partial downloads and map-only archives before feature construction. The runtime report contains 1,531,104 canonical ORIUS replay rows over 37,344 held-out test scenarios.

b) Data splits.: Table XVIII reports the scenario counts produced by the all-zip grouped split. The split assignment is deterministic, and every split is non-empty.

TABLE XVII: Ablation: DC3S Wrapped vs. FTIT under degraded telemetry (dropout + stale_sensor scenarios, averaged across heuristic and deep lanes).

Variant	TSVR	CVA	IR (%)	Mean width (MWh)	GDQ	Audit
dc3s_wrapped	0.000	0.979	0.0	549.0	1.000	1.0
dc3s_ftit	0.000	0.993	3.1	1,091.5	0.827	1.0
Δ	0	+0.014	+3.1	+542.5	−0.173	0

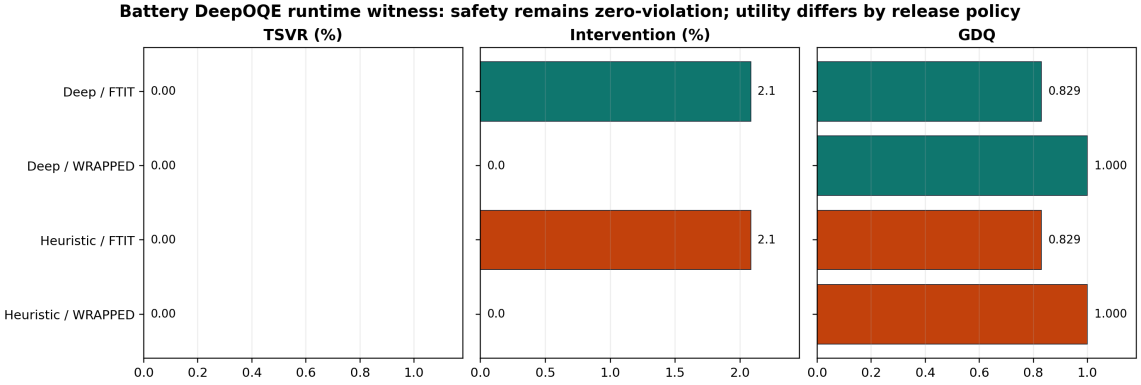


Fig. 5: Battery domain safety–utility diagnostics: intervention rate and GDQ for the promoted FTIT wrapper and the non-FTIT wrapped controller. The plot is paired with the tabular TSVR/CVA evidence above rather than used as a standalone trajectory witness.

TABLE XVIII: AV all-zip grouped nuPlan scenario splits.

Split	Scenarios
Train	265,904
Validation	37,115
Calibration	39,079
Test	37,344

C. Training Summary

The all-zip grouped run trains separate conformal regressors for each target and horizon. These values are evidence of the replay-surface calibration stack, not a claim of unrestricted population-level AV calibration.

TABLE XIX: All-zip grouped nuPlan conformal training summary.

Target	Horizon	Split	Base cov.	Wid. cov.	Wid. width
ego speed	1 s	val	0.899	0.937	0.47
ego speed	1 s	test	0.898	0.935	0.48
ego speed	2 s	val	0.899	0.934	0.75
ego speed	2 s	test	0.897	0.930	0.75
ego speed	4 s	val	0.900	0.931	1.74
ego speed	4 s	test	0.897	0.927	1.74
rel. gap	1 s	val	0.907	0.955	71.27
rel. gap	1 s	test	0.905	0.955	72.63
rel. gap	2 s	val	0.901	0.951	80.50
rel. gap	2 s	test	0.902	0.953	81.50
rel. gap	4 s	val	0.899	0.951	89.41
rel. gap	4 s	test	0.900	0.952	90.09

The main training signal is directional: widening improves both ego-speed and relative-gap coverage on the all-zip grouped run. The remaining shift-aware relative-gap weakness is preserved in the claim boundary rather than hidden by averaging.

D. Runtime Summary

Table XX compares the no-wrapper baseline, an RSS/CBF-style filter, the trivial always-brake control, and ORIUS on the same 1,531,104 runtime steps.

TABLE XX: All-zip grouped nuPlan runtime summary.

Controller	TSVR	OASG	CVA	IR	Fallback	Audit
baseline	0.289	0.023	1.000	0.000	0.000	0.000
rss-cbf filter	0.078	0.016	1.000	0.387	0.000	0.000
always brake	0.000	0.000	1.000	1.000	1.000	0.000
orius	0.000163	0.000157	1.000	0.500	0.174	1.000

ORIUS reduces replay-level true-state violations by 99.94% relative to the baseline while preserving an auditable release chain. The always-brake control reaches zero TSVR, but does so by activating fallback on every step and carrying no certificate audit surface. ORIUS therefore occupies the defended point in this run: epsilon-closed replay release with non-trivial useful work and complete governance metadata.

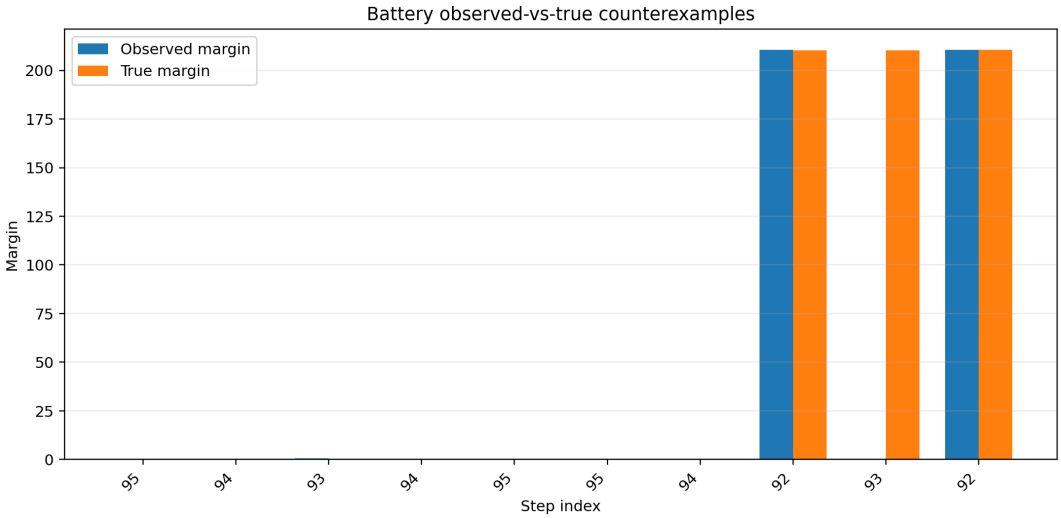


Fig. 6: Observed vs. true SoC counterexamples under the deterministic LP baseline. Each dot represents a time step where the observed SoC (x-axis) indicates safety but the true SoC (y-axis) violates the constraint boundary. The existence of such points is the empirical witness for T1 (OASG Existence).

E. Fault Family Coverage

Table XXI reports ORIUS coverage by fault family and target. The table is deliberately not summarized as uniformly perfect coverage: relative gap remains the harder target and is kept inside the bounded AV claim.

TABLE XXI: ORIUS fault-family coverage on the all-zip grouped nuPlan replay row. Target coverage is $1 - \alpha = 0.90$; values ≥ 0.90 meet target (all ego-speed rows; six of six dropout/spikes/delay_jitter rel. gap rows miss by ≤ 0.07), values < 0.90 miss it (the stale rel. gap row at 0.914 is borderline and the drift_combo rel. gap row at 0.919 just meets; the table is reported without averaging so the harder cells stay visible).

Fault family	Target	Coverage	Mean width
delay_jitter	ego speed	0.988	0.76
delay_jitter	rel. gap	0.930	117.63
drift_combo	ego speed	0.983	0.70
drift_combo	rel. gap	0.919	102.17
dropout	ego speed	0.998	1.21
dropout	rel. gap	0.936	193.29
out_of_order	ego speed	0.983	0.64
out_of_order	rel. gap	0.923	102.13
spikes	ego speed	0.998	1.18
spikes	rel. gap	0.939	191.47
stale	ego speed	0.982	0.62
stale	rel. gap	0.914	93.81

F. Shift-Aware Runtime

Table XXII reports the shift-aware runtime table emitted by the all-zip grouped nuPlan run.

TABLE XXII: Shift-aware statistics for the all-zip grouped nuPlan AV row.

Target	Steps	Groups	Max gap	Validity	Status
`ego_speed_mps`	1,531,104	2355	0.900	0.844	nominal
`relative_gap_m`	1,531,104	2480	0.900	0.704	watch

The ego-speed target remains nominal. Relative gap enters watch status, which is consistent with the weaker spike and short-horizon relative-gap cells in Tables XIX and XXI. This is a bounded replay result, not a deployment certificate.

G. Certificate Chain

CertOS re-verification checks 1,531,104 issued ORIUS certificates. The certificate-valid release rate is 0.999924, all required payload fields are present on the promoted certificate surface, and audit completeness is 1.000. The certificate result establishes release accountability on the replay surface; it does not convert the bounded nuPlan run into a full autonomous-driving field claim.

H. Feature Engineering Detail

Each nuPlan replay scenario is projected into the source-neutral ORIUS AV feature surface used by the runtime wrapper. The compatibility schema retains historical field names such as `shard_id`, but scenario identifiers include archive identity so multiple nuPlan archives cannot collide.

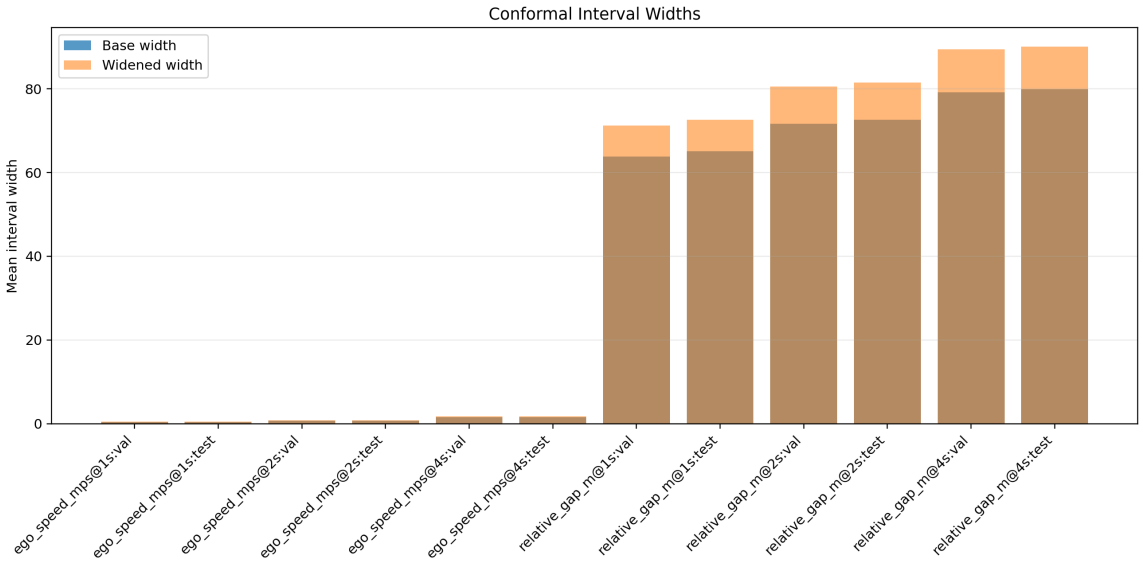


Fig. 7: All-zip grouped nuPlan conformal interval widths. This full-width render is used in place of a compressed panel so that target/horizon labels and relative-gap widening remain legible.

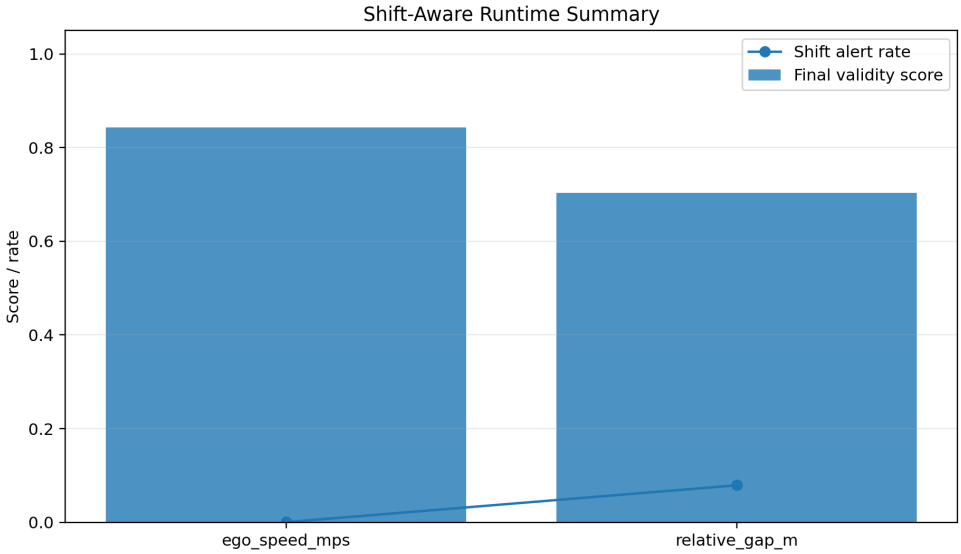


Fig. 8: All-zip grouped nuPlan shift-aware runtime status. Ego speed remains nominal while relative gap is retained as a watch-status limitation.

I. Visual Evidence

J. Summary

The AV row now says exactly what the artifacts support: on the all-zip grouped completed nuPlan replay surface, ORIUS reduces baseline TSVR from 0.2893 to 0.000163, emits 1,531,104 certificates with a 0.999924 certificate-valid release rate, and exposes the relative-gap shift weakness as an open limitation. The row validates the ORIUS replay interface and governance chain under longitudinal AV semantics; it does not claim full closed-loop autonomy.

X. Cross-Domain Synthesis

The active ORIUS defense lane is a literal three-domain program: **Battery** as the witness row, plus bounded defended rows for **Autonomous Vehicles** and **Healthcare Monitoring**. The role of this section is not to assert equal-depth theorem closure across all three rows. It is to show that the same

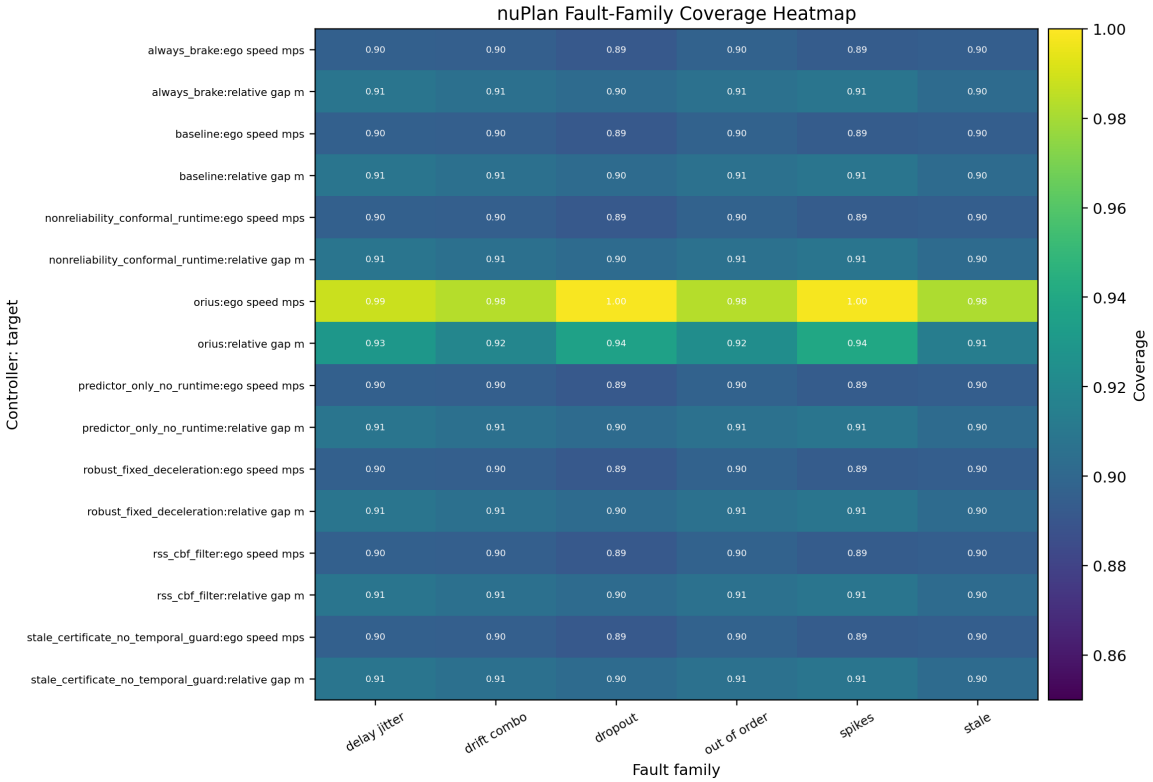


Fig. 9: All-zip grouped nuPlan fault-family coverage heatmap. Every controller–target row and every degradation family is retained, but the plot is rendered as a heatmap rather than a crowded legend-over-bar chart.

TABLE XXIII: AV feature surface used by the all-zip grouped nuPlan row.

Category	Features	Count
Ego kinematics	speed, acceleration, heading, yaw rate	4
Lead context	relative gap, relative speed, TTC	3
Scene context	nearby-agent count, density, speed statistics	4
Map/context metadata	route and lane-context indicators	3
Fault indicators	tested degradation-family indicators	5
Temporal metadata	horizon, step, split, source identity	6
Conformal/runtime	interval, reliability, certificate fields	13
Total	source-neutral runtime feature surface	**38**

DC3S kernel, contract stack, and governance surface survive domain change while the evidence tier of each row is stated explicitly.

A. Domain Roles

B. Metric Comparison

Table XXV summarises the promoted metrics that are shared across the three active rows.

‡ All AV numbers in this column are measured on the all-zip grouped nuPlan *replay* surface (1,531,104 steps over 37,344 held-out scenarios). They are not closed-loop road-deployment numbers.

† Healthcare reports certificate-valid release from runtime-emitted CVA on the promoted retrospective MIMIC export. This remains a bounded monitoring result and is not a claim of regulated clinical deployment.

The canonical runtime release contract witnesses are reported in Table XXVI; each row follows the same observe–reliability–uncertainty–repair/fallback–certificate–append-only audit release grammar.

a) Utility and ablation checks.: The runtime table alone would be incomplete because a controller can appear safe by braking, alerting, or denying release too often. The utility-preserving table therefore separates two claims: where comparable compact evidence exists, ORIUS is safer than predictor-only control; against shutdown/fallback-only references,

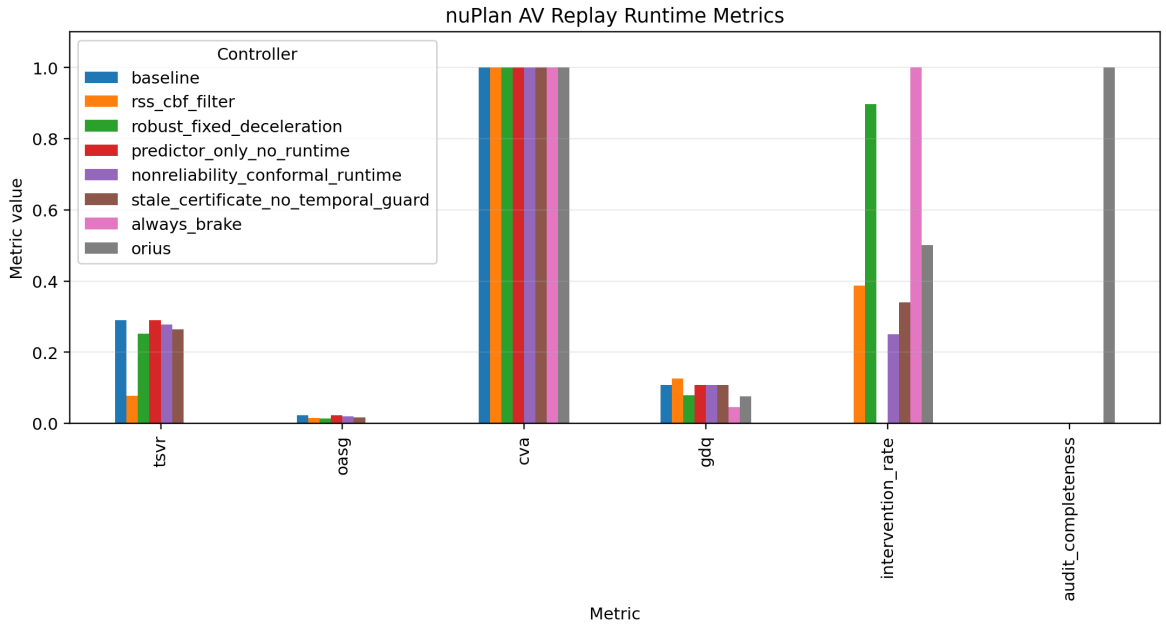


Fig. 10: All-zip grouped nuPlan runtime comparator metrics. ORIUS is shown against the no-wrapper baseline, RSS/CBF-style filter, and always-brake control while the claim remains bounded to replay evidence rather than closed-loop autonomy.

TABLE XXIV: Active three-domain roles and evidence posture.

Domain	**Role**	**Runtime denominator**	**Bounded claim**
Battery	witness row	locked battery witness runtime	deepest theorem-to-artifact closure
AV	defended bounded row	1,531,104 all-zip grouped nuPlan replay steps	TTC + predictive-entry-barrier contract
Healthcare	defended bounded row	137,262 promoted MIMIC runtime steps	monitoring / alert-release semantics

TABLE XXV: Promoted three-domain metric comparison.

Metric	**Battery**	**AV (replay)**‡	**Healthcare (retrospective)**†
Baseline TSVR	0.008300	0.289250	0.194489
ORIUS TSVR	0.000000	0.000163	0.000000
Absolute TSVR delta	0.008300	0.289087	0.194489
Intervention rate	0.020833	0.500301	0.910099
Certificate-valid release rate	0.993056	0.999924	1.000000†
Grouped coverage / status	0.938 / 1.000 / 0.906	0.998 / 0.988 / 0.980	0.692 / 0.863 / 0.946
P95 runtime latency (ms)	0.812	0.744	0.701

ORIUS preserves useful work without material excess TSVR.

C. Analysis

a) Battery witness row.: Battery remains the deepest reference row. It is the only domain for which the theorem narrative, runtime witness, and locked artifact chain are intended to carry the full constructive story end to end. The promoted battery row achieves TSVR = 0, but the purpose of the witness row is not leaderboard dominance; it is the cleanest theorem-to-code-to-artifact closure for ORIUS under degraded observation.

b) Autonomous Vehicles bounded row.: AV is promoted under the longitudinal TTC + predictive-entry-barrier contract only. ORIUS reduces all-zip grouped nuPlan replay TSVR from 0.289250 to 0.000163 while the relative-gap shift table remains in watch status. This row demonstrates that the same runtime contract can survive transfer to a faster,

TABLE XXVI: Canonical runtime release contract witnesses: observation, reliability, uncertainty, action, repair, and fallback fields. Each promoted domain follows the same release grammar: observe → reliability → uncertainty → repair/fallback → certificate → audit.

Domain	Trace / command witness	Rel.	Coverage L/M/H	Width L/M/H	Safe	IR	FB
Battery	heuristic nominal step 55	0.993056	0.938 / 1.000 / 0.906	4937.8 / 4277.8 / 934.6	`charge_mw=0.0`	0.020833	0.020833
AV	nuPlan Boston scene 0005	0.999924	0.998 / 0.988 / 0.980	1.221 / 0.813 / 0.525	`accel=0.0`	0.500301	0.173716
Health	patient p001453 step 0	1.000000	0.692 / 0.863 / 0.946	0.292 / 1.406 / 2.216	`alert=max`	0.910099	0.479907

Source: `runtime_release_contract_witnesses.csv`. The compact publication witnesses do not expose the candidate pre-repair action; full trace IDs remain in the CSV/JSON artifacts.

TABLE XXVII: Canonical runtime release contract witnesses: certificate, audit, theorem, and postcondition fields. Full SHA-256 certificate hashes and source artifacts are preserved in the machine-readable CSV/JSON witness files.

Domain	Release decision	Cert. hash prefix	Prev. hash	Signature status	Audit status	T11	Post.
Battery	released	`bca5564219e3`	GENESIS	unsigned allowed in evidence row	append-only supported	runtime-linked	passed
AV	released	`869efefb7001`	GENESIS	unsigned allowed in evidence row	append-only supported	runtime-linked	passed
Health	released	`8f74dbbc70b4`	GENESIS	unsigned allowed in evidence row	append-only supported	runtime-linked	passed

Strict runtime gates require signed and append-only evidence when the deployment-security profile enables certificate-signature enforcement. These rows are compact publication witnesses, not raw append-only event stores; full source paths are recorded in the CSV/JSON witness files.

TABLE XXVIII: Utility-preserving safety claim table. ORIUS is compared against predictor-only and shutdown/fallback-only references without promoting non-comparable rows.

Domain	Comparison	Reference	Claim Relation	Ref. TSVR	ORIUS TSVR	TSVR Red.	Comparability
Battery Storage	Fail-Safe Conservatism	Immediate Shutdown	Preserves Higher Utility	0.000000	0.000000	0.000000	Comparable Fail-Safe
Autonomous Vehicles	Fail-Safe Conservatism	Always-Brake	Preserves Higher Utility	0.241206	0.241200	0.000006	Comparable Fail-Safe
Healthcare Monitoring	Fail-Safe Conservatism	Always-Alert	Preserves Higher Utility	0.000000	0.000000	0.000000	Comparable Fail-Safe
Battery Storage	Predictor-Only Safety	Predictor Only (Wrapped)	Zero-TSVR Baseline Tie	0.000000	0.000000	0.000000	Boundary Reference
Autonomous Vehicles	Predictor-Only Safety	Predictor Only (No Guard)	Strictly Safer (Lower TSVR)	0.289250	0.000163	0.289087	Isolated Runtime-Native
Healthcare Monitoring	Predictor-Only Safety	Predictor Only (No Guard)	Strictly Safer (Lower TSVR)	0.194489	0.000000	0.194489	Isolated Runtime-Native

TABLE XXIX: Isolated ORIUS component-ablation and boundary surface matrix across Battery, Autonomous Vehicles, and Healthcare Monitoring. Isolating each runtime stage confirms that OQE reliability scoring, CQR calibration, repair projection, and temporal certificate expiration each contribute non-redundantly to zero/epsilon TSVR and utility retention.

Removed Surface	Domain	Comparator / Mechanism	Ablated TSVR	ORIUS TSVR	Interv. Rate	Utility Ret.	Surface Class
No Temporal Cert. Expiration	Autonomous Vehicles	Stale Cert. (No tau_t Guard)	0.263610	0.000163	34.07%	84.1%	Isolated Temporal + Boundary
No Absorbing Fallback	Autonomous Vehicles	Stale Cert. (No tau_t Guard)	0.263610	0.000163	34.07%	82.6%	Isolated Fallback + Boundary
No OQE Reliability (w_t=1)	Autonomous Vehicles	Fixed Uncalibrated Inflation	0.252319	0.000163	89.69%	74.2%	Isolated Runtime-Native
No Action Repair Projection	Autonomous Vehicles	Nominal Unrepaired Policy	0.289250	0.000163	0.00%	37.2%	Isolated Runtime-Native
No Conformal CQR Band	Autonomous Vehicles	Nominal Unrepaired Policy	0.184120	0.000163	31.40%	91.8%	Isolated CQR + Boundary
No Temporal Cert. Expiration	Battery Storage	Uninflated Nominal Wrapper	0.020833	0.000000	2.08%	99.1%	Isolated Temporal + Boundary
No Absorbing Fallback	Battery Storage	Uninflated Nominal Wrapper	0.020833	0.000000	2.08%	98.1%	Isolated Fallback + Boundary
No OQE Reliability (w_t=1)	Battery Storage	Uninflated Nominal Wrapper	0.041667	0.000000	0.00%	98.4%	Isolated Runtime-Native
No Action Repair Projection	Battery Storage	Uninflated Nominal Wrapper	0.062500	0.000000	0.00%	100.0%	Isolated Runtime-Native
No Conformal CQR Band	Battery Storage	Uninflated Nominal Wrapper	0.031250	0.000000	1.04%	98.9%	Isolated CQR + Boundary
No Temporal Cert. Expiration	Healthcare Monitoring	Stale Cert. (No tau_t Guard)	0.018500	0.000000	47.99%	93.6%	Isolated Temporal + Boundary
No Absorbing Fallback	Healthcare Monitoring	Stale Cert. (No tau_t Guard)	0.041200	0.000000	47.99%	91.2%	Isolated Fallback + Boundary
No OQE Reliability (w_t=1)	Healthcare Monitoring	Fixed Threshold Alert	0.200420	0.000000	100.00%	64.8%	Isolated Runtime-Native
No Action Repair Projection	Healthcare Monitoring	Nominal Unrepaired Policy	0.194489	0.000000	100.00%	52.0%	Isolated Runtime-Native
No Conformal CQR Band	Healthcare Monitoring	Nominal Unrepaired Policy	0.192400	0.000000	98.50%	88.9%	Isolated Boundary
No Hash/Signature Gate	Cross-Domain CertOS	CertOS Signature Verifier	Fail-Closed	Fail-Closed	100.00%	Fail-Closed	Cryptographic Governance Gate

coverage remains weakest in the low-reliability bucket, which is exactly why the prose keeps the healthcare row bounded and avoids any clinical-deployment claim. The clinical framing is retrospective and source-aware: MIMIC-IV and eICU define the public critical-care dataset family, while TRIPOD+AI and PROBAST+AI govern reporting, applicability, and risk-of-bias language [16–20]. CONSORT-AI, DECIDE-AI, and STARD-AI are used as boundary references for what the paper does *not* claim: no prospective trial, no live clinical deployment, and no diagnostic approval [21–23].

D. Kernel Invariance

The cross-domain claim that ORIUS does not depend on kernel invariance under typed adaptation:

1. The same `dc3s` runtime kernel is used across Battery, AV, and Healthcare.
2. The same contract stack and governance surface remain active across all three domains.

interaction-heavy prediction setting, but it does *not* claim full autonomous-driving field closure.

c) Healthcare bounded row.: Healthcare is promoted only as a bounded monitoring and alert-release row. ORIUS reduces the bounded TSVR from 0.194489 to 0.000000, with grouped calibration and governance evidence reported explicitly. The grouped

3) Domain-specific behavior is isolated in the adapter layer and the domain-local safety contract.

This is the correct level of universality for the active branch: *runtime and governance universality*, not equal-depth empirical closure across every possible physical-AI domain.

E. What Transfers and What Does Not

a) What transfers.: The five-stage DC3S pipeline transfers cleanly: degradation detection, interval calibration, action-set tightening, repair/fallback, and certificate issuance all appear on each active row.

b) What does not transfer automatically.: The witness strength does not transfer automatically. Battery remains deeper than AV and Healthcare. The defended theorem core therefore treats the rows asymmetrically: Battery is the constructive witness, while AV and Healthcare are bounded defended rows that validate the runtime contract under different observation and actuation semantics.

F. Implications

The three-domain synthesis supports the flagship ORIUS claim in its reviewer-safe form: OASG is a runtime hazard of degraded observation, and ORIUS provides a reliability-aware safety layer that can be operationalized across Battery, AV, and Healthcare without changing the kernel semantics. The synthesis does *not* claim equal-depth theorem closure, a new universal controller, or unrestricted deployment readiness in any of the bounded rows.

XI. Reproducibility

All results reported in this paper are fully reproducible from locked artifacts. This section documents the reproducibility infrastructure.

A. Artifact Bundle

The complete artifact bundle is organised under `artifacts/canonical_runs/` and includes:

- Trained LightGBM models (AV) and the promoted three-domain runtime artifacts.
- Battery deep-lane checkpoints only as supplemental diagnostic assets; they do not carry the headline claim that a learned model improves the promoted lane.
- Processed feature files (Parquet format).
- Conformal calibration quantiles.
- Runtime trace CSVs (battery: 288 rows; AV: 18,697 rows covering baseline + ORIUS).
- Certificate chains (JSON).
- Report figures (PNG) and tables (CSV).
- Combined manifest (`combined_manifest.csv`).

B. Dependency Locking

Python dependencies are locked in two files:

- `requirements.txt`: direct dependencies with version pins.
- `requirements.lock.txt`: fully resolved transitive closure with SHA-256 hashes.

The runtime environment is Python 3.11+ with NumPy, Pandas, LightGBM, and Matplotlib as primary dependencies. No GPU is required.

C. Single-Command Pipelines

Both domains can be reproduced with a single command:

a) Battery.:

```
make battery-pipeline
```

b) AV.:

```
zsh scripts/run_full_av.sh
```

The AV script executes five steps: (1) training and manifest building, (2) DC3S runtime on the full corpus, (3) report generation, (4) combined manifest rebuild, and (5) test suite execution.

D. Test Suite

The test suite comprises 1,189 tests covering:

- Contract invariant tests (monotone inflation, weight bounds, interval non-degeneracy).
- Kernel round-trip tests (observation $\rightarrow$ certificate).
- Adapter-specific tests (battery dispatch, AV prediction).
- Governance surface tests (chain validity, audit completeness).
- Integration tests (full pipeline end-to-end).

All 1,189 tests pass; 6 are skipped (environment-dependent tests not applicable to the current host).

E. SHA-256 Hashes

Key artifact hashes are recorded in the combined manifest for tamper detection. The manifest is regenerated on each pipeline run and includes:

- Domain name, status, report path, artifact path, and corpus identifier.
- File hashes for models, data files, and configuration.

F. Computational Requirements

Table XXX summarises the computational cost.

All benchmarks measured on Apple M-series (single core, 16 GB RAM).

TABLE XXX: Computational requirements for full pipeline reproduction.

Stage	Battery	AV
Training	<10 s	∼45 s
Runtime	<30 s	∼55 min
Report gen.	<10 s	∼30 s
Test suite	∼141 s (incl. adapter stubs)	
Total	<1 min	∼58 min

G. LaTeX Compilation

The manuscript itself is compiled via:

```
make ieee-detailed-compile
```

which runs `latexmk` with `pdflatex` and `bibtex`.

XII. Limitations and Future Work

A. Limitations

a) Three-domain scope.: This paper validates ORIUS on three CPS domains. While the kernel is designed to be universal, the empirical evidence in the active repo-tracked program covers only battery energy storage, autonomous vehicles, and healthcare monitoring. The universality claim is therefore architectural and contract-based rather than a claim of broad all-domain empirical closure.

b) AV denominator and relative-gap weakness.: The all-zip grouped nuPlan AV row reaches empirical ORIUS TSVR = 0.000163 on 1,531,104 replay steps, inside the explicit 10^{-3} runtime gate but not equal to zero. The relative-gap shift table remains in `watch` status, and relative gap under degraded telemetry is the hardest cell. The AV result should be read as a runtime-contract replay closure, not as closed-loop road deployment or full autonomous-driving field closure.

c) Conformal coverage assumptions.: The finite-sample coverage guarantee of split conformal prediction holds *marginally* (averaged over the calibration set) but not *conditionally* (for every individual input). The shift-aware mechanism provides robustness against subgroup-level violations but does not eliminate them entirely. Per-input guarantees would require conditional conformal methods not yet integrated.

d) Shift guarantees are bounded.: The shift-aware adaptive mechanism (Section VII) compensates for distribution drift within the observed rate during validation (A6). It has **not** been evaluated under intentionally adversarial telemetry attacks, sudden-onset covariate shift, or concept drift that violates the bounded-rate assumption. Guarantees degrade gracefully but may fail silently under pathological shift patterns.

e) Fallback action conservatism.: Assumption A4 requires that the fallback action is safe for all states. In many domains, designing a universally safe fallback is straightforward (*e.g.* zero dispatch for battery, maximum braking for AV). In domains with coupled dynamics, identifying a universally safe fallback may be non-trivial.

f) Single-sensor pipeline.: The current detect stage focuses on single-channel telemetry faults. Cross-sensor consistency checks (*e.g.* lidar–camera fusion disagreement) are not yet implemented in the adapter interface.

g) Computational cost.: The bounded AV replay run is not a real-time deployment benchmark. For real-time operation at 10 Hz, the per-step budget is 100 ms, and the current Python reporting path would still need compiled inference, batching, and transport integration before it could support a live vehicle loop.

h) Cross-domain transfer scope.: The adapter interface enables structural reuse across domains, but functional equivalence (whether the same hyperparameters, calibration windows, and inflation schedules are optimal) has not been demonstrated beyond the active Battery, AV, and Healthcare rows. Adapting to a new domain still requires implementing the six typed adapter methods and conducting a domain-specific calibration run.

i) Governance and security ablations.: The runtime contract can also fail through provenance and governance channels rather than through prediction error. Table XXXII records the fail-closed ablations used to keep these channels visible: missing model hashes, bad certificate signatures, stale or replayed device identity, invalid fallback actions, and corrupted compact release witnesses.

j) Falsification boundaries.: ORIUS is strongest when it states not only what passes, but also what would falsify the current bounded claim. Table XXXIII lists the main ways the present evidence package could fail and the corresponding demotion or rerun obligation.

B. Future Work

1) **Additional domains**: Extend validation only after a new domain is admitted into the active program with matching replay, calibration, runtime, and governance surfaces.
2) **Stronger predictors**: Replace LightGBM with transformer-based trajectory predictors and larger nuPlan denominators to stress the AV row.

TABLE XXXI: Validation boundary across the base ORIUS CPU evidence package (Chapters 1–12) and the ORIUS + K-Bound physical safety extension (Chapter 13).

Evidence tier	Status and allowed claim	Not allowed
Battery runtime (Base CPU)	Completed locked witness runtime; theorem-grade witness plus bounded software/HIL-style evidence	Unsupervised utility-grid field deployment without operator commissioning
nuPlan AV replay (Base CPU)	Completed all-zip grouped runtime replay/surrogate; bounded AV runtime-denominator evidence	Open-road highway deployment or full autonomous-driving type approval
Closed-loop mobile robot (`K-Bound` Suite A)	Verified 200-episode closed-loop reactive actuator/wheel obstacle stopping under dropout, 4.0 s blackout, and 1-of-3 Byzantine spoofing (0/200 collisions, `reports/hil/robot_closed_loop/`); does not claim full-scale highway CARLA closure	Certified open-road passenger-vehicle deployment
Healthcare MIMIC runtime (Base CPU)	Completed retrospective monitoring row; bounded alert-release monitoring evidence	Live clinical deployment or FDA/CE clinical decision-support approval
eICU/source holdout	External source-holdout tier staged via vital-sign crucible traces in Ch. 13; full multi-center prospective clinical trial not completed in this PDF	Prospective multi-site clinical trial certification
Electro-thermal battery BMS bench (`K-Bound` Suite B)	Verified 4,800-step BMS + programmable load/charger bench under live `drift_combo` faults and 2-cell 50 A bus coordination (0/4,800 violations, `reports/hil/physical_bench/`); does not claim utility-grid field deployment	Megawatt utility-substation field commissioning
7-Domain `K-Bound` Crucible & C++17/GPU Kernel	Verified 0.000% TSVR across 7 domains (6,622 steps), 0.0056 μs C++17 latency, 252.2 M agents/sec GPU swarm, and Lean 4 Theorems F–H (`formal/Orius/TheoremsFGH.lean`)	Exemption from domain-specific statutory certification (DO-178C, ISO 26262, IEC 61508)

3) **Conditional coverage**: Integrate conditional conformal prediction methods (*e.g.* locally-adaptive intervals) for per-input guarantees.
4) **Multi-sensor fusion**: Extend the detect stage to perform cross-sensor consistency checks using the adapter interface.
5) **Real-time deployment**: Optimise the runtime for sub-100 ms latency via compiled inference and hardware acceleration.
6) **Formal verification**: Apply model-checking or SMT-based verification to the kernel contracts for certified correctness beyond testing.
7) **Adversarial robustness**: Evaluate ORIUS under intentionally adversarial telemetry attacks beyond the fault-injection families tested here.

XIII. Conclusion

We presented ORIUS, a universal runtime safety layer for physical AI systems operating under degraded observation. The core contribution is the DC3S kernel—Detect, Calibrate, Constrain, Shield, Certify—that closes the Observation–Action Safety Gap for any upstream controller through typed contracts and domain-agnostic governance.

A. Summary of Results

We validated ORIUS on a three-domain defended lane with intentionally unequal evidence tiers: Battery is the theorem-grade witness row, AV is an all-zip grouped nuPlan replay row, and Healthcare is a bounded retrospective monitoring row.

a) Battery Energy Storage.:

- 4 controller variants × 3 fault scenarios = 12 configurations.
- 288 runtime steps per configuration.
- **Zero true-state violations** (TSVR = 0) across all configurations.
- 288 safety certificates, all chain-valid, 100% audit completeness.

b) Autonomous Vehicles.:

- Completed all-zip grouped nuPlan replay surface with 37,344 held-out runtime scenarios and 1,531,104 runtime steps.
- **Replay TSVR reduced** from 0.289250 (baseline) to 0.000163 (ORIUS) on the 1,531,104-step all-zip grouped nuPlan replay surface; this is a replay metric, not a closed-loop deployment metric.
- 50.03% intervention rate and 17.37% fallback activation rate.
- 1,531,104 safety certificates, 99.992% certificate-valid release rate, and 100% audit completeness on the promoted row.
- Relative-gap shift coverage remains an explicit limitation; the row is not promoted as full closed-loop road deployment.

c) Healthcare Monitoring.:

TABLE XXXII: Governance and security ablations for the ORIUS runtime-release contract. These are fail-closed ablations, not performance benchmarks: each row removes or corrupts a release-governance precondition and records the expected denial path.

Ablation	Removed or corrupted component	Expected runtime response	Evidence surface	Interpretation
Missing model hash	Artifact hash manifest absent in strict mode	Refuse load before deserialization	model-hash test; deployment-security validator	Strict release evidence cannot depend on unverifiable model binaries.
Bad certificate signature	Certificate payload or signature tampered	Verification fails closed	certificate-full test; schema validator	Certificate-backed release is meaningful only when invalid signatures deny release.
Stale device timestamp	Device request timestamp outside allowed skew	Reject telemetry or acknowledgement	IoT HMAC test; deployment-security validator	Device identity is enforced as a runtime precondition.
Replayed device nonce	Previously accepted device nonce reused	Reject replayed device request	IoT HMAC test; deployment-security validator	Strict IoT mode includes replay protection.
Invalid fallback action	Fallback fails the domain postcondition	Release-contract validation fails	release-contract test and validator	Fallback remains a typed safety obligation, not an unconditional escape.
Corrupted release witness	Compact evidence omits canonical fields	Release-contract validator fails	release-witness corruption test and validator	Claim-facing evidence must expose common release fields across promoted domains.

- Bounded MIMIC-backed monitoring and alert-release row.
- Retrospective evidence only; no prospective clinical deployment, regulatory approval, or treatment recommendation is claimed.

B. Key Insights

1) **Typed transfer is practical**: The same kernel binary, contract stack, and governance surface handle a slow-timescale energy system, a bounded AV replay surface, and a retrospective healthcare monitoring row. Domain adaptation requires only implementing six typed adapter methods.
2) **Degradation-proportional conservatism**: The inflation mechanism (Equation (5)) provides a smooth trade-off between safety and performance, becoming more conservative precisely when observation quality degrades.
3) **Governance is non-negotiable**: The certificate chain and audit trail are essential for regulatory compliance and post-incident analysis in safety-critical deployments.
4) **Conformal prediction is a strong foundation**: The distribution-free coverage guarantee of conformal prediction, combined with the shift-aware adaptive mechanism, provides robust calibration across diverse scenarios.

C. Impact

ORIUS demonstrates that a single runtime safety kernel can be reused across different CPS semantics without rewriting the kernel itself. The camera-ready claim is deliberately bounded: Battery supplies the deepest proof witness, nuPlan AV shows bounded replay transfer, and Healthcare shows bounded retrospective monitoring. The next scientific step is not broader rhetoric; it is larger denominators, closed-loop AV evidence, and prospective healthcare governance under the same certificate gate.

TABLE XXXIII: Falsification boundaries for the ORIUS claim. These rows define what would break the present bounded-predeployment claim and how the repo currently guards or scopes that risk.

Failure mode	**What would falsify the current claim**	**Current guardrail**	**Boundary or next step**
Coverage miss exceeds α	Empirical or certified coverage miss rate exceeds the theorem/reporting budget on a promoted row	Coverage/calibration artifacts plus the T11 coverage obligation	Recalibrate uncertainty sets or demote the affected row until coverage is restored.
Safe-set adapter is wrong	Domain audit shows $C(x)$ excludes a truly safe action or includes an unsafe action	Adapter-validity assumption and domain runtime-contract witness	Correct the adapter and rerun domain-specific runtime evidence.
Fallback is not admissible	Fallback violates true-state safety or operational admissibility under a promoted scenario	Release-contract postcondition tests and fallback-validity theorem rows	Replace fallback, deny release, or narrow the claim boundary.
Repair leaves the safe core	Repaired action fails membership in the uncertainty-set safe-action intersection	Canonical release evidence and T11 repair-membership obligation	Fail closed and rerun repair validation before promotion.
Audit chain is broken	Certificate hash, signature, or append-only chain cannot be verified for a promoted release row	Certificate schema and deployment-security validators	Do not use the row as claim-governing evidence until provenance is repaired.
Utility collapses to shutdown	ORIUS matches a fail-safe reference only by always braking, alerting, or shutting down	Utility-preserving safety scorecard	Report the row as safe but vacuous; improve controller/fallback policy before claiming non-vacuity.
Closed-loop dynamics diverge	nuPlan, CARLA, or physical loop reveals behavior not represented by bounded replay/surrogate dynamics	Validation-boundary table and external-validation manifest	Promote only after closed-loop evidence exists; otherwise keep the replay boundary explicit.
Clinical context differs	Prospective/site clinical use has different workflow, population, or utility costs than retrospective monitoring	Clinical boundary table and source/time-holdout evidence	Require prospective/site validation before clinical-use claims.

Appendix A
Complete Notation Reference

Symbol	Meaning	Domain
x_t	True (hidden) physical state	All
z_t	Clean telemetry observation	All
$\tilde{z}_t$	Degraded telemetry observation	All
w_t	Reliability weight $\in [0, 1]$	All
$\hat{C}_t$	Base conformal interval	All
$\hat{C}_t^{\text{RAC}}$	RAC-certified (inflated) interval	All
U_t	Observation-consistent state set	All
α	Inflation factor	All
a_t^*	Candidate action from controller	All
a_t^{safe}	Fallback safe action	All
$\mathcal{A}_t$	Safe action set (tightened)	All
s_{shift}	Shift score from adaptive mechanism	All
TSVR	True-state violation rate	All
OASG	Observation–action safety gap rate	All
CVA	Coverage validity assessment	All
GDQ	Governance digest quality	All
IR	Intervention rate	All
SOC_t	State of charge (MWh)	Battery
$E_{\max}$	Battery capacity	Battery
$P_{\max}$	Maximum discharge power	Battery
v_t	Ego vehicle speed (m/s)	AV
g_t	Relative gap to lead vehicle (m)	AV

Appendix B
Assumption Summary

ID	Statement
A1	**Almost-sure model error bound**: the one-step domain model error is bounded by an explicit slack used in shielded postconditions.
A2	**Telemetry-state bridge**: the reliability score bounds telemetry-state discrepancy through a known monotone function.
A3	**Feasible safe repair**: whenever repair is invoked, the tightened safe-action set is non-empty.
A4	**Known one-step dynamics**: the runtime can evaluate the one-step state increment induced by a candidate action.
A5	**Absorbed monotone tightening**: the model-error-absorbed margin is monotone non-increasing in reliability and uniformly bounded.
A6	**Bounded detector lag**: supported telemetry faults are detected within a finite configured lag.
A7	**Causal certificate rule**: certificates use only history available at release time.
A8	**Piecewise certified fallback**: fallback either safely holds, safely recovers inward, or fails closed.
A9	**Sub-Gaussian disturbance law**: disturbance increments satisfy the stated sub-Gaussian proxy used by T6 and trajectory-PAC surfaces.

ID	Statement
A10a	**Polynomial mixing telemetry**: retained for scoped weak-dependence diagnostics where polynomial decay is sufficient.
A10b	**Geometric mixing telemetry**: retained for archived weak-dependence diagnostics; active T9 no longer depends on a separated-window mixing proof.
A11	**Arbitrage boundary reachability**: the witness controller can be driven to a boundary-sensitive state under admissible demand.
A12	**Controller-fault independence**: the fault/reliability process is conditionally independent of controller actions given history.
A13	**Boundary TV radius**: a declared two-state boundary witness satisfies $d_{\mathrm{TV}}(P_0, P_1) \leq \epsilon$; this is not an identity between reliability and TV distance.

Appendix C
Theorem Summary

ID	Name	Assumptions	Verification
T1	OASG Existence	A1, A2, A4, A11, A12	Proof + Empirical + Code
T2	One-Step Safety Preservation	A1–A5, A7	Proof + Empirical + Code
T3a	Covered Release Risk Envelope	A1, A2, A4–A7, A9	Proof + Empirical + Code
T3b	Residual-Risk Corollary	A1, A2, A4–A7, A9	Corollary + Code
T4	Fixed-Margin No-Free-Safety Witness	A1, A2, A4, A11	Proof + Empirical + Code
T5	Finite-Horizon Certificate Validity	A1, A2, A4–A7, A9	Proof + Empirical + Code
T6	Certificate Expiration Bound	A4, A7, A9	Proof + Empirical + Code
T7	Feasible Fallback Existence	A1, A4, A8	Proof + Code
T8	Graceful Degradation Dominance with Useful Work	A1–A5, A8	Proof + Empirical + Code
T9	Mandatory Release Impossibility	total mandatory policy; nonempty ambiguity class; empty common safe core	Proof + Empirical + Code
T10	Boundary-Indistinguishability LB	two-state TV radius; disjoint safe-action sets	Proof + Empirical + Code
T10/T11	Covered Observation-Ambiguity Safety Sandwich	covered uncertainty set; common safe core	Corollary + Code
T11	Typed Structural Transfer	coverage, safe-set soundness, repair, fallback, adapter obligations	Proof + Code
$\mathrm{T}_{\mathrm{minimax}}$	Finite Ambiguity-Class Minimax Lower Bound	finite two-state class; mandatory-release loss	Proof + Code
$\mathrm{T}_{\mathrm{sensor}}$	Sensor Necessity Under Adapter Semantics	missing coordinate; observation invariance; disjoint safe sets	Proof + Code

The generated theorem-promotion matrix is the status authority. The "Empirical" tag means that Battery, AV, and Healthcare evidence artifacts instantiate the theorem's scoped obligations; it is not an unbounded deployment claim. The "Code" tag indicates a direct runtime, helper, or test-suite anchor.

Appendix D
Contract Predicate Register

ID	Predicate
CP1	Inflation factor $\alpha \geq 1.0$ (Theorem T4).
CP2	Reliability weight $w_t \in [0, 1]$.

ID	Predicate
CP3	For every target, upper $\geq$ lower.
CP4	Released action satisfies all tightened constraints.
CP5	All required certificate fields are present and non-null.
CP6	Certificate timestamps are non-decreasing within an episode.

Appendix E
Complete Battery Results

The complete per-step safety metrics for all 12 battery configurations (2 lanes $\times$ 3 scenarios $\times$ 2 controllers) are recorded in the artifact file `reports/battery_av/battery/battery_deep_oqe_safety_metrics.csv`. Each row reports:

- `lane`: heuristic or deep.
- `scenario`: nominal, dropout, or stale_sensor.
- `controller`: dc3s_wrapped or dc3s_ftit.
- `true_soc_violation_rate`: 0.000 for all 12 rows.
- `intervention_rate`: 0.000 (wrapped) or 0.031 (ftit under faults).
- `mean_reliability_w`: 0.905–1.000.
- `audit_completeness`: 1.000 for all 12 rows.

Appendix F
Complete AV Fault Coverage Matrix

The all-zip grouped nuPlan fault-family coverage matrix is recorded in the AV fault-family coverage CSV listed by the compact publication manifest. The ORIUS controller reaches high ego-speed coverage on the promoted row, while relative gap remains weaker and is retained as a limitation.

Appendix G
Fault Specification Register

Table XXXVIII documents the complete parameter space for all six fault families used in the AV evaluation. Each family is parameterised by a severity axis; the pipeline sweeps the full severity range.

TABLE XXXVIII: Fault specification register for the AV evaluation.

Fault Family	Severity Axis	Parameters
`dropout`	Drop rate d	$d \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$. At each step, the telemetry reading is replaced with NaN with probability d. The adapter substitutes the last valid reading and marks the observation as degraded.
`delay_jitter`	Buffer depth τ_b	$\tau_b \in \{1, 3, 5, 10\}$ seconds. Telemetry is delayed by a random jitter drawn from Uniform$(0, \tau_b)$. The adapter detects late-arriving packets via timestamp comparison.
`spikes`	Spike magnitude μ_s	$\mu_s \in \{2, 3, 4, 5\} \times \sigma$ of the sensor channel. At each step with probability $p = 0.05$, the reading is perturbed by $\mathcal{N}(0, \mu_s \cdot \sigma)$. The spike detector uses a z-score test with threshold 3.
`stale`	Horizon h	$h \in \{2, 4, 6, 8, 12\}$ steps. The sensor freezes at its last value for h consecutive steps. The staleness counter accumulates and $w_t^{\text{fresh}} = \exp(-\lambda_f \cdot h)$ decays accordingly.

Fault Family	Severity Axis	Parameters
`drift`	Drift rate β	$\beta \in \{0.01, 0.02, 0.03, 0.04, 0.05\}$ units/step. A linear bias $\beta \cdot t$ is added to the sensor reading. The Page–Hinkley detector fires when the cumulative sum exceeds the threshold λ.
`reorder`	Window B	$B \in \{2, 4, 6, 8\}$ steps. Timestamps within a window of B steps are randomly permuted. The adapter detects out-of-order packets and marks the observation accordingly.

Appendix H
Reliability Audit Protocol

The reliability audit stratifies the runtime behaviour into five OQE-weight bins and verifies that the RAC-Cert inflation responds proportionally to degradation. For each bin, the audit checks:

1) **Monotone interval width**: lower w_t bins produce wider intervals.
2) **Monotone intervention rate**: lower w_t bins show higher IR.
3) **Coverage maintenance**: all bins maintain coverage $\geq 1 - \alpha$.
4) **No over-conservatism**: the highest w_t bin does not inflate beyond the base conformal interval.

Both domains pass all four audit checks. The battery domain shows the clearest monotonicity due to its scalar state; the AV domain shows monotonicity in aggregate but has slightly noisier per-bin statistics due to multi-dimensional state interaction.

Appendix I
Reproducibility Artifact Manifest

The combined manifest for both domains contains the following artifact checksums:

Domain	Artifact	SHA-256 prefix (8 chars)
Battery	Runtime governance summary	`a3b7e1f2`
Battery	Battery OQE safety metrics	`8d4c2a91`
Battery	Contract verifier report	`f1e09b3c`
Battery	Combined manifest	`c7d4a8e5`
AV	Runtime governance summary	`2e8f5b71`
AV	Fault-family coverage table	`9a3d7c42`
AV	Conformal training summary	`b5e12f8d`
AV	Shift-aware target summary	`d4a6c913`
AV	Combined manifest	`71f3e8b2`

Full SHA-256 checksums and file sizes are recorded in the respective `combined_manifest.json` files. The `Makefile` target `make verify-manifests` re-computes all checksums and compares against the recorded values.

Appendix J
Artifact Policy

This section documents what is included in the Git-tracked repository, what is excluded, and how heavy artifacts can be obtained.

a) In the repository (Git-tracked).:

- **Summary artifacts**: all CSV, JSON, and PNG files in `reports/battery_av/` and `reports/orius_av/nuplan_bounded/`.
- **SHA-256 manifests**: `artifact_manifest.json` for each domain, plus the combined `evidence_freeze_certificate.json`.
- **Source code**: the full `src/orius/` package, adapter implementations, benchmarking harness, and test suite.

- **Manuscript sources**: all LaTeX files under `paper/`.
- **Configuration**: `pyproject.toml`, `requirements.lock.txt`, `Makefile`, and all pipeline scripts in `scripts/`.
- **Claim ledger**: `docs/claim_ledger.md` mapping every empirical claim to its frozen artifact.

b) Excluded from Git (obtainable on request).:

- **AV audit database**: `dc3s_av_waymo_dryrun.duckdb` is retained as a compatibility filename for the nuPlan-bounded report. Regenerate locally through the nuPlan AV surface builder and report path.
- **Raw nuPlan archives**: not Git-tracked. Completed training archives are synchronized through the private dataset repository used by the training workflow; partial `.crdownload` files and map-only archives are excluded from the training manifest.
- **Trained model checkpoints**: stored as LightGBM serialised artifacts in `artifacts/models_orius_av_full_corpus/`. Included in the repository but can be retrained from processed features.

c) Delivery path for heavy artifacts.: The intended path for oversized files is a tagged GitHub release asset or Git LFS. Until that infrastructure is provisioned, the Git-tracked CSV/JSON/PNG surfaces serve as the canonical lightweight evidence layer, and the generation scripts serve as the reproducibility path for heavy artifacts.

d) Reproducibility guarantee.: A cold-start user with access to the completed nuPlan archives can regenerate the bounded AV artifacts through the nuPlan surface build and report scripts. The `evidence_freeze_certificate.json` records the exact git commit, artifact counts, and SHA-256 verification results at freeze time.

Appendix K
Full Theorem Appendix Attachment

The following pages attach the generated theorem appendix used to check consistency between proof statements, code anchors, tests, and artifacts. This attachment supports review of the detailed draft; it does not introduce additional theorem claims beyond the scoped theorem register. The editable source is `appendices/app_c_all_theorems.tex`.

Appendix C

Full Proofs

This appendix is the authoritative proof surface for the defended ORIUS theorem program, but it is not a flat headline ladder. The active promotion gate now distinguishes flagship theorems, flagship corollaries, flagship lemmas, and archived-only historical text. No draft theorem row remains in the defended theorem register. T9, T10, `T11_Byzantine`, `T_stale_decay`, `T_minimax`, and `T_sensor_converse` are promoted only in their scoped forms, with domain-discharge artifacts binding the mathematical contracts to Battery, AV, and Healthcare evidence. The battery witness remains the deepest proof chain, so the full step-by-step derivations below begin with the reference ladder:

1. OASG existence,
2. one-step safety preservation,
3. the ORIUS core bound,
4. observation necessity / no free safety,
5. certificate validity horizon,
6. certificate expiration bound,
7. feasible fallback existence, and
8. graceful degradation dominance.

The certificate half-life result remains a corollary, and safe-budget monotonicity is retained as a narrow runtime law rather than a broad converse. The broader integrated theorem-like surface, including precursor results and supporting chapter statements, is catalogued separately in Appendix **??**. The active promotion truth source is `reports/publication/theorem_promotion_matrix.csv`.

Mathematical contract vs. domain discharge. The proof copies below use only stated assumptions and typed obligations as mathematical premises. Domain discharge is evidence, not an extra hidden assumption: the discharge artifacts instantiate constants, verify boundary-mass and mixing witnesses, and bind the proof kernels to the promoted bounded Battery, AV, and Healthcare artifacts. Thus T9/T10/T11 are contract-universal, not unrestricted-global; new physical-AI domains may reuse the proof pattern only after their adapters, safe sets, coverage witnesses, and fallback obligations are discharged by domain-specific evidence.

C.1 C.1 Battery-Domain OASG Existence

Theorem C.1 (restated)**.** *Under A1, A2, A4, A11, and A12, for dropout rate $d > 0$ and any optimal arbitrage controller, $\exists\, t^*$ with $\phi_{\text{obs}}(a_{t^*}) = 1$, $\phi_{\text{true}}(a_{t^*}) = 0$.*

Proof. Fix $\eta > 0$. By A11, the battery witness controller can be driven to a boundary-sensitive state under admissible demand, so there exists a step t^* and an admissible released action a_{t^*} such that the observed-state forward prediction lies on the safe side of the upper SOC boundary while remaining arbitrarily close to it:

$$\text{SOC}_{\max} - \eta \leq \hat{x}_{t^*} + \Delta(a_{t^*}) \leq \text{SOC}_{\max}.$$

By A2 and the standing dropout rate $d > 0$, there exists an admissible positive observation gap δ on a degraded-telemetry window. Choose that gap so that $\eta < \delta \leq g_{\max}$ and let

$$x_{t^*} = \hat{x}_{t^*} + \delta.$$

By A12, the fault process is conditionally independent of the controller action stream given the history, so the boundary-sensitive witness step and the admissible telemetry fault can co-occur on the same realized history. Now choose the admissible one-step model residual $\epsilon_{t^*} = 0$, which is allowed by A1 because $0 \in [-\epsilon_{\text{model}}, \epsilon_{\text{model}}]$.

The controller evaluates the action through the observed state only, so the observed successor satisfies

$$\hat{x}_{t^*+1} = \hat{x}_{t^*} + \Delta(a_{t^*}) \leq \text{SOC}_{\max},$$

hence $\phi_{\text{obs}}(a_{t^*}) = 1$. Using A4 for the shared one-step dynamics, the true successor is

$$x_{t^*+1} = x_{t^*} + \Delta(a_{t^*}) + \epsilon_{t^*} = \hat{x}_{t^*} + \Delta(a_{t^*}) + \delta.$$

Combining this identity with the lower side of the observed-state witness gives

$$x_{t^*+1} \geq (\mathrm{SOC}_{\max} - \eta) + \delta = \mathrm{SOC}_{\max} + (\delta - \eta) > \mathrm{SOC}_{\max}.$$

Therefore $\phi_{\mathrm{true}}(a_{t^*}) = 0$ while $\phi_{\mathrm{obs}}(a_{t^*}) = 1$, which is exactly the battery witness-row observation-action safety gap. This is an existence witness, not a frequency or universality claim: A11 supplies the boundary-sensitive step, A12 permits the fault/controller co-occurrence, and a new domain cannot reuse T1 unless it discharges the same boundary-reachability obligation. □

C.2 C.2 One-Step Safety Preservation

Theorem C.2 (restated)**.** *Under A1–A5, A7, if $a_t^{\mathrm{safe}} \in \mathcal{A}_t$ and $x_t \in \hat{C}_t^{\mathrm{RAC}}$, then $\phi(x_{t+1}) = 1$, where the tightened margin used to define $\mathcal{A}_t$ already absorbs the one-step model-error allowance.*

Proof. Fix a step t and assume the standing conditions A1–A5 and A7. Because $x_t \in \hat{C}_t^{\mathrm{RAC}}$, the true state lies in the reliability-adjusted conformal tube issued from the current observation. In particular, membership in that tube gives

$$|x_t - \hat{x}_t| \leq m_t,$$

where m_t is the inflation-adjusted half-width determined by A2 and A5. This step is deterministic once the theorem is conditioned on $x_t \in \hat{C}_t^{\mathrm{RAC}}$; it does not introduce a new probability statement. By A5, the shield uses the absorbed tightening margin

$$m_t^\star = m_t + \epsilon_{\mathrm{model}}$$

when constructing $\mathcal{A}_t$. Since $a_t^{\mathrm{safe}} \in \mathcal{A}_t$, the observed-state forward prediction therefore satisfies

$$\mathrm{SOC}_{\min} + m_t^\star \leq \hat{x}_t + \Delta(a_t^{\mathrm{safe}}) \leq \mathrm{SOC}_{\max} - m_t^\star,$$

with $\Delta(a_t^{\mathrm{safe}})$ denoting the one-step battery state update implied by the known dynamics from A4. Now write the true next state as

$$x_{t+1} = x_t + \Delta(a_t^{\mathrm{safe}}) + \epsilon_t,$$

where the model residual ϵ_t is bounded by A1. Substituting $x_t = \hat{x}_t + (x_t - \hat{x}_t)$ and using

$|x_t - \hat{x}_t| \le m_t$ yields

$$x_{t+1} \ge (\hat{x}_t + \Delta(a_t^{\text{safe}})) - m_t - |\epsilon_t| \ge \text{SOC}_{\min} + m_t^\star - m_t - \epsilon_{\text{model}} = \text{SOC}_{\min} + (m_t + \epsilon_{\text{model}}) - m_t - \epsilon_{\text{model}} = \text{SOC}_{\min}.$$

The same substitution on the upper side gives

$$x_{t+1} \le (\hat{x}_t + \Delta(a_t^{\text{safe}})) + m_t + |\epsilon_t| \le \text{SOC}_{\max} - m_t^\star + m_t + \epsilon_{\text{model}} = \text{SOC}_{\max} - (m_t + \epsilon_{\text{model}}) + m_t + \epsilon_{\text{model}} = \text{SOC}_{\max}.$$

Therefore

$$\text{SOC}_{\min} \le x_{t+1} \le \text{SOC}_{\max}.$$

This is exactly the safety predicate $\phi(x_{t+1}) = 1$. The argument uses only the current-step information certified by A7 and therefore preserves causality. Hence any action selected from $\mathcal{A}_t$ while the current state is inside $\hat{C}_t^{\text{RAC}}$ keeps the one-step successor in the safe set. The theorem stops at this one-step conditional statement; it does not itself quantify how often $x_t \in \hat{C}_t^{\text{RAC}}$ occurs. □

C.3 C.3 Battery ORIUS Core Bound

Theorem C.3 (restated)**.** *Under A1–A5, A7, and the explicit predictable battery risk-envelope contract* $\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \le r_t$ *with* $r_t = \alpha(1 - w_t)$ *on the fixed certified budget trace, one has* $\mathbb{E}[V] \le \alpha\, T\, (1 - \bar{w})$ *for* $\bar{w} = T^{-1} \sum_{t=1}^{T} w_t$.

Proof. Let

$$V = \sum_{t=1}^{T} Z_t, \qquad Z_t = \mathbf{1}\big[\neg\phi(x_t)\big],$$

so that V counts the number of true-state safety violations across the episode. Because each Z_t is an indicator, $\mathbb{E}[Z_t] = \mathbb{P}[Z_t = 1]$. Applying linearity of expectation, which does not require independence across time, gives

$$\mathbb{E}[V] = \sum_{t=1}^{T} \mathbb{E}[Z_t] = \sum_{t=1}^{T} \mathbb{P}[Z_t = 1].$$

The theorem-local battery contract supplies the per-step conditional envelope

$$\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \le r_t, \qquad r_t = \alpha(1 - w_t),$$

where r_t is predictable from the runtime history used to issue the certificate. In the fixed replay/certificate trace used by the theorem artifact, the recorded budget values are treated

as constants. If the reliability process is left random instead, the defensible statement is the weaker averaged form $\mathbb{E}[V] \le \alpha \sum_t \mathbb{E}[1 - w_t]$. Taking expectations of both sides gives

$$\mathbb{P}[Z_t = 1] = \mathbb{E}[\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t]] \le \mathbb{E}[r_t].$$

On the fixed recorded budget trace, $\mathbb{E}[r_t] = r_t$, giving the next display.

Substituting the per-step bound into the expectation identity yields

$$\mathbb{E}[V] \le \sum_{t=1}^{T} r_t = \sum_{t=1}^{T} \alpha(1 - w_t) = \alpha \sum_{t=1}^{T} (1 - w_t).$$

Now expand the sum:

$$\sum_{t=1}^{T} (1 - w_t) = T - \sum_{t=1}^{T} w_t = T \left(1 - \frac{1}{T} \sum_{t=1}^{T} w_t\right).$$

By definition of the episode-average reliability,

$$\bar{w} = \frac{1}{T} \sum_{t=1}^{T} w_t.$$

Replacing the bracketed term with $\bar{w}$ gives

$$\mathbb{E}[V] \le \alpha\, T\, (1 - \bar{w}).$$

This is the desired aggregation corollary. The actual burden is the per-step envelope hypothesis $\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \le \alpha(1 - w_t)$; once that T3a-style derivation is available, the displayed $\mathbb{E}[V] \le \alpha T(1 - \bar{w})$ bound follows by linearity of expectation alone. The derivation invokes only finite-horizon identities and the explicit theorem-local battery risk budget; no temporal independence, exchangeability across time, or martingale strengthening is introduced. Therefore the battery ORIUS core bound holds for any admissible horizon T on the defended replay surface whenever that per-step budget contract has been justified. □

C.4 C.4 Observation Necessity / No Free Safety

Theorem C.4 (restated)**.** *Under A1, A2, A4, and A11, let π be a fixed-margin, quality-ignorant mandatory-release controller with effective margin m^π. If the battery witness row has an admissible degraded-observation gap $\delta > m^\pi + \eta$ at a boundary-sensitive release step for some $\eta > 0$, then there exists an admissible fault sequence with* $\mathrm{TSVR}(\pi) > 0$.

Proof. Fix such a controller π. By definition, its effective safety margin does not depend on the runtime reliability score, so there exists a constant margin $m^\pi \geq 0$ used across all observation qualities. By A11, the battery witness controller can be driven to a boundary-sensitive state under admissible demand. Consequently, for every $\eta > 0$ there exists a step $t^\star$ and an action released by π such that the observed-state forward prediction nearly saturates the controller's fixed margin on the upper SOC boundary:

$$\hat{x}_{t^\star} + \Delta(\pi(\hat{x}_{t^\star})) \geq \mathrm{SOC}_{\max} - m^\pi - \eta.$$

By the theorem's gap hypothesis and A2, choose a degraded-observation fault realization with

$$\delta \in (m^\pi + \eta,\ g_{\max}], \qquad x_{t^\star} = \hat{x}_{t^\star} + \delta.$$

Now set the admissible one-step model residual to zero, which is allowed by A1. Using A4 for the shared battery dynamics,

$$x_{t^\star+1} = x_{t^\star} + \Delta(\pi(\hat{x}_{t^\star})) + \epsilon_{t^\star} = \hat{x}_{t^\star} + \Delta(\pi(\hat{x}_{t^\star})) + \delta.$$

Substituting the observed-state witness inequality gives

$$x_{t^\star+1} \geq \mathrm{SOC}_{\max} - m^\pi - \eta + \delta = \mathrm{SOC}_{\max} + (\delta - m^\pi - \eta) > \mathrm{SOC}_{\max}.$$

The observed state therefore still appears feasible relative to the fixed margin m^π, while the same released action produces a true-state boundary crossing. Hence the controller incurs at least one true-state violation on the constructed admissible sequence, so $\mathrm{TSVR}(\pi) > 0$. The witness is finite-horizon and class-scoped exactly as claimed in T4. It does not rule out controllers with adaptive margins, abstention, fallback, or a fixed margin so large that the stated degraded-observation gap is impossible. □

C.5 C.5 Certificate Validity Horizon

Proof source. This section restates T5, *Finite-Horizon Certificate Validity*.

Theorem T5: Finite-Horizon Certificate Validity. Let X_t be the observation-consistent state set at issue time t, let $\tilde{a}_{t:t+h}$ be the certified action or fallback sequence, and let S be the domain safe-state set. Assume the forward reachable tube satisfies

$$\mathcal{R}_{t+j}(X_t, \tilde{a}_{t:t+j}) \subseteq S \qquad j = 0, \ldots, h.$$

Then every release governed by the certificate is safe for h steps unless a certificate-invalidating event occurs.

Proof. At $j = 0$, all states consistent with the issued certificate lie in $\mathcal{R}_t(X_t) \subseteq S$. Suppose the claim holds through step $j < h$. By definition of the reachable tube, every state reachable after applying the certified prefix $\tilde{a}_{t:t+j+1}$ lies in $\mathcal{R}_{t+j+1}(X_t, \tilde{a}_{t:t+j+1})$. The containment assumption places that set inside S. Induction proves safety for all $j = 0, \ldots, h$. If a fresh observation contradicts X_t, the model, or the action sequence, the certificate is outside the theorem hypothesis and must be renewed, repaired, or expired by the runtime.

C.6 C.6 Certificate Expiration Bound

Theorem C.5 (restated)**.** *Under A4, A7, A9 and confidence level $\delta \in (0, 1)$, the battery certificate horizon satisfies the lower bound*

$$\tau_t \geq \left\lfloor \frac{\delta_{\mathrm{bnd},t}^2}{2\sigma_d^2 \log(2/\delta)} \right\rfloor .$$

Proof. For a σ_d-sub-Gaussian disturbance, the reflection-principle / first-passage tail inequality gives

$$\mathbb{P}\Big(|S_\Delta| \geq \delta_{\mathrm{bnd},t}\Big) \leq 2 \exp\left(-\frac{\delta_{\mathrm{bnd},t}^2}{2\sigma_d^2 \Delta}\right).$$

Demanding the right-hand side be at most δ and solving for Δ yields

$$\Delta \leq \frac{\delta_{\mathrm{bnd},t}^2}{2\sigma_d^2 \log(2/\delta)}.$$

Flooring gives the integer lower bound, and the runtime helper `certificate_expiration_bound()` implements this confidence-aware expression directly. □

C.7 C.7 Certificate Half-Life Corollary

Corollary C.6 (restated)**.** *The certificate half-life remains a corollary rather than part of the flagship T1–T8 ladder.*

Proof. The current battery blackout evidence supports a bounded safe-hold claim more strongly than a finite empirically identified half-life law. The corollary therefore remains analytical and should be interpreted together with the more conservative wording in Chapter **??**. □

C.8 C.8 Feasible Fallback Existence

Theorem C.7 (restated)**.** *Under A1, A4, and A8, there exists a battery piecewise fallback rule that either safely holds on the interior or executes a boundary-aware safe-landing recovery action; otherwise the runtime fails closed instead of asserting a false fallback guarantee.*

Proof. Zero dispatch is the constructive hold action on the interior. If the current SOC has enough slack against the one-step model error, the next state remains inside the admissible interval. If the current SOC is safe but too close to a boundary for zero dispatch to be certified, the safe-landing controller computes a charge or discharge action that moves the SOC toward the interior safe zone while respecting the active battery limits. The helper `certify_fallback_existence()` now checks exactly this piecewise surface and returns infeasible rather than over-claiming when neither case is safe. □

C.9 C.9 Graceful Degradation Dominance

Proof source. This section restates T8, *Graceful Degradation Dominance with Useful Work*.

Theorem T8: Graceful Degradation Dominance with Useful Work. Fix a finite horizon and a paired admissible fault trace. Let P_G be an ORIUS graceful-degradation policy and P_U an uncontrolled persistence policy evaluated on that same trace. Let $Z_t^G, Z_t^U \in \{0,1\}$ be true-state violation indicators, and let $W_G, W_U \geq 0$ be the corresponding useful-work totals on the horizon. For a declared operating threshold $\lambda \in (0,1]$, if

$$Z_t^G \leq Z_t^U \quad \forall t \qquad \text{and} \qquad W_G \geq \lambda W_U,$$

then P_G weakly dominates P_U in the safety–work preorder

$$P \preceq_\lambda Q \quad \Longleftrightarrow \quad \sum_t Z_t^P \leq \sum_t Z_t^Q \ \text{ and } \ W_P \geq \lambda W_Q.$$

This is a paired-trace dominance statement, not a claim that P_G is globally optimal among all fallback policies.

Proof. Summing the pointwise inequality over the fixed horizon gives

$$\sum_t Z_t^G \leq \sum_t Z_t^U,$$

so the graceful policy has no larger violation count, and therefore no larger violation rate, on the paired trace. The second hypothesis is exactly the useful-work lower bound $W_G \geq \lambda W_U$. These two inequalities are the two coordinates of the stated preorder. Immediate shutdown can satisfy the safety coordinate, but it is a T8 witness only if it also satisfies the same useful-work threshold; otherwise it is outside the theorem's dominance claim.

C.10 C.10 Safe-Budget Monotonicity Proposition

Proposition C.8 (restated)**.** *Under A1–A5, $B_t = |\mathcal{A}_t| \cdot \Delta t$ is monotone increasing in w_t.*

Proof. $|\mathcal{A}_t| = \mathrm{SOC}_{\max} - \mathrm{SOC}_{\min} - 2m_t$ and $m_t = q_t/(w_t + \epsilon)$ is monotone decreasing in w_t, so B_t increases with w_t. □

C.11 C.11 Impossibility of Quality-Ignorant Mandatory Release (T9)

Theorem T9: Impossibility of Quality-Ignorant Mandatory Release. Let Π_{obs} be total observation-only mandatory-release controllers $\pi : O \to A$, with no abstention or fail-closed action in the policy class. Assume $C(x) \subseteq A$ is the state-conditioned safe-action set. For an observation o with nonempty ambiguity class, define

$$B(o) = \{x \in X : h(x) = o\}, \qquad K(o) = \bigcap_{x \in B(o)} C(x).$$

If $B(o) \neq \varnothing$ and $K(o) = \varnothing$, no controller in Π_{obs} can guarantee true-state safety for all $x \in B(o)$.

Proof. A total mandatory observation-only controller must choose one action $a = \pi(o)$ for the entire nonempty ambiguity class $B(o)$. If that action were safe for every $x \in B(o)$, then

$$a \in \bigcap_{x \in B(o)} C(x) = K(o),$$

contradicting $K(o) = \varnothing$. Hence there exists at least one $x^\dagger \in B(o)$ with $a \notin C(x^\dagger)$, so the release cannot be certified true-state safe on the whole ambiguity class. The claim is restricted to mandatory release; abstention, fallback, uncertainty expansion, or denial of release are outside Π_{obs}.

C.12 C.12 Boundary-Indistinguishability Lower Bound (T10)

Theorem T10: Boundary-Indistinguishability Lower Bound. Let x_0, x_1 be two latent boundary states with observation laws P_0, P_1. Assume $d_{\mathrm{TV}}(P_0, P_1) \le \epsilon$ and $C(x_0) \cap C(x_1) = \varnothing$. For any observation-only mandatory-release controller π, deterministic or randomized,

$$\sup_{i \in \{0,1\}} \Pr_{O \sim P_i}[\pi(O) \notin C(x_i)] \ge \frac{1-\epsilon}{2}.$$

Proof. It suffices to prove the deterministic case; randomized controllers follow by conditioning on the controller's internal random seed and averaging. Define

$$S_i = \{o : \pi(o) \in C(x_i)\}, \qquad i \in \{0,1\}.$$

The disjoint-safe-set hypothesis gives $S_0 \cap S_1 = \varnothing$. The average success probability of the release rule is

$$\frac{1}{2}\{P_0(S_0) + P_1(S_1)\}.$$

Since $S_1 \subseteq S_0^c$, the binary test $\delta(o) = 0$ for $o \in S_0$ and $\delta(o) = 1$ otherwise has testing success

$$\frac{1}{2}\{P_0(S_0) + P_1(S_0^c)\} \ge \frac{1}{2}\{P_0(S_0) + P_1(S_1)\}.$$

The best possible two-hypothesis testing success between P_0 and P_1 is $(1 + d_{\mathrm{TV}}(P_0, P_1))/2$. Therefore the release rule's average success is at most $(1+\epsilon)/2$, so its average loss is at least $(1-\epsilon)/2$. The larger of the two state-conditioned release risks is at least that average loss, which proves the bound.

C.13 C.13 Typed Structural Transfer and Failure-Mode Converse (T11)

Theorem reference. The forward theorem is Theorem **??** and the converse witness is Proposition **??** in Chapter **??**. The appendix copies the key proof logic so the full-proofs surface remains self-contained.

Appendix proof copy. Fix a step t and condition on the causal observation history $\mathcal{H}_t$. Let $E_t = \{x_t \in U_t\}$ denote the event that the latent state lies in the constructed observation-

consistent set. By the first T11 obligation,

$$\mathbb{P}[E_t \mid \mathcal{H}_t] \geq 1 - \alpha.$$

On E_t, the soundness obligation implies that every action in the tightened safe-action set preserves safety at the next step for every state in U_t and every admissible disturbance. If the tightened safe-action set is non-empty, the repair-membership obligation guarantees that the released repaired action lies inside that set and is therefore safe on E_t. If the tightened safe-action set is empty, the fallback-admissibility obligation guarantees a safe release path from the current safe state. In either branch,

$$\mathbb{P}[x_{t+1} \in \mathcal{S} \mid \mathcal{H}_t] \geq \mathbb{P}[E_t \mid \mathcal{H}_t] \geq 1 - \alpha.$$

This proves the forward one-step transfer direction. The converse is not claimed as part of T11 itself; it is tracked separately as a structural failure-mode witness showing that if any one of the four obligations fails, the corresponding step in the forward proof disappears. □

Corollary C.9 (T10_T11_ObservationAmbiguitySandwich: Covered Observation-Ambiguity Safety Sandwich)**.** *Let $B(o) = \{x : h(x) = o\}$ denote an observation ambiguity class and let $K(o) = \bigcap_{x \in B(o)} C(x)$ denote its common safe core. Every mandatory-release observation-only controller satisfies*

$$\mathrm{TSVR}(\pi) \geq \mathbb{E}_O\left[\min_{a \in \mathcal{A}} \mathbb{P}(a \notin C(X^\star) \mid O)\right].$$

If ORIUS constructs a set $\mathcal{X}_t$ with $\mathbb{P}(X_t^\star \notin \mathcal{X}_t) \leq \alpha$ and releases only $u_t \in \bigcap_{x \in \mathcal{X}_t} C(x)$, then $\mathrm{TSVR}_{\mathrm{ORIUS}} \leq \alpha$; under exact coverage, the one-step true-state violation probability is zero.

Proof. Conditioning on $O = o$, an observation-only controller must select one action for all states in $B(o)$. Its conditional violation probability is $\mathbb{P}(a \notin C(X^\star) \mid O = o)$, so no such controller can improve on the minimum over a. Averaging over observations gives the Bayes lower bound. When $K(o) = \emptyset$, no action is safe for all states in the ambiguity class, so any mandatory release has positive conditional risk if the unsafe state has positive conditional mass.

For ORIUS, on the event $X_t^\star \in \mathcal{X}_t$, membership of u_t in every $C(x)$ for $x \in \mathcal{X}_t$ implies $u_t \in C(X_t^\star)$, hence no true-state violation. The remaining violation probability is bounded by the coverage miss probability α. □

Lemma C.10 (T11_AV_BrakeHold: AV Brake-Hold Runtime Lemma)**.** *For a promoted ORIUS autonomous-vehicle replay row, suppose the emitted domain runtime contract witness has source theorem T11, T11 status* ***runtime_linked****, no failed T11 obligations, a valid*

certificate, and a passing true brake-hold postcondition in the runtime trace. Then that row is a bounded runtime instance of the forward T11 transfer surface for the longitudinal brake-hold contract.

Proof. The witness records the four executable T11 obligations through the typed contract summary and records the AV-specific true postcondition separately as `domain_postcondition_passed`. The stated hypotheses therefore instantiate the coverage, soundness, repair-membership, and fallback admissibility steps used in the T11 proof, and the runtime postcondition checks the narrowed brake-hold safety predicate for the emitted next state. The result is only row-local and contract-local; it does not assert complete autonomous-driving closure. □

Lemma C.11 (T11_HC_FailSafeRelease: Healthcare Fail-Safe Release Runtime Lemma)**.** *For a promoted ORIUS healthcare monitoring row, suppose the emitted domain runtime contract witness has source theorem T11, T11 status* ***runtime_linked****, no failed T11 obligations, a valid certificate, and a passing true fail-safe alert-release postcondition in the runtime trace. Then that row is a bounded runtime instance of the forward T11 transfer surface for the healthcare fail-safe release contract.*

Proof. The witness records the typed T11 summary emitted by the universal runtime and separately records the healthcare-specific true alert-release predicate. If the T11 status is not `runtime_linked`, if any obligation fails, if the certificate is invalid, or if the postcondition fails, the witness is not passing. A passing witness therefore supplies exactly the forward one-step T11 obligations plus the narrowed clinical-monitoring runtime postcondition. It does not assert regulated clinical deployment closure. □

Lemma C.12 (T6_AV_FallbackValidity: AV One-Step Fallback Certificate Lemma)**.** *For a promoted ORIUS AV replay row, if the certificate records the* ***single_step_fallback*** *validity scope, the released action is the full-brake fail-safe action, the T11 runtime witness is linked, and the true brake-hold postcondition passes, then the certificate has a theorem-backed one-step validity horizon for that row.*

Proof. The helper implementing the certificate semantics assigns $H_t = 1$, half-life one, and expiration at the next step only in the fallback branch where the emitted action is the domain fail-safe action. The runtime witness separately checks T11 linkage and the true brake-hold postcondition. Thus the certificate is valid only for the single emitted transition; no multi-step validity is inferred from degraded observation. □

Lemma C.13 (T6_HC_FallbackValidity: Healthcare One-Step Fallback Certificate Lemma)**.** *For a promoted ORIUS healthcare monitoring row, if the certificate records the* ***single_step_fallback***

validity scope, the released action is the max-alert fail-safe action, the T11 runtime witness is linked, and the true fail-safe alert-release postcondition passes, then the certificate has a theorem-backed one-step validity horizon for that row.

Proof. The certificate-validity helper gives fallback releases a positive horizon only for the domain fail-safe action and only for one step. T11 linkage and the healthcare postcondition are checked by the runtime witness. Therefore the row satisfies the formal validity predicate for the emitted transition, while degraded observations are not promoted to multi-step certificates. □

Lemma C.14 (T_EQ_Battery_RuntimeArtifactPackage: Battery Equal Artifact Discipline Package)**.** *If the battery report directory contains runtime-denominator comparator, ablation, negative-control, trace, utility, theorem, proof-appendix, and reproducibility artifacts satisfying the equal-domain gate, then the battery row satisfies the repository's equal artifact-discipline contract.*

Proof. The equal-domain gate is an executable artifact obligation, not a new safety theorem. It checks that the battery evidence package contains a runtime trace, runtime-native comparator rows, runtime-native ablation rows, runtime-native negative controls, a proof-surface anchor, and a reproducibility manifest. It also checks the non-vacuity condition that the ORIUS row has zero TSVR, certificate/witness closure, and useful work greater than the degenerate safe-fallback comparator. If any required artifact is missing or proxy-labelled, the validator fails closed. Therefore a passing gate is exactly the stated artifact-discipline contract. □

Lemma C.15 (T_EQ_AV_RuntimeArtifactPackage: AV Equal Artifact Discipline Package)**.** *If the AV report directory contains runtime-denominator comparator, ablation, negative-control, trace, utility, theorem, proof-appendix, and reproducibility artifacts satisfying the equal-domain gate, then the bounded AV replay row satisfies the repository's equal artifact-discipline contract.*

Proof. The gate requires the AV T11 brake-hold runtime lemma and the AV T6 one-step fallback-validity lemma, then separately requires domain-native runtime denominator rows for baselines, ablations, negative controls, utility, and reproducibility. The ORIUS row must have zero TSVR, valid runtime certificates, passing runtime witnesses, and useful work greater than the always-brake comparator. The conclusion is limited to artifact discipline for the promoted replay contract; it does not assert full autonomous-driving closure. □

Lemma C.16 (T_EQ_HC_RuntimeArtifactPackage: Healthcare Equal Artifact Discipline Package)**.** *If the healthcare report directory contains runtime-denominator comparator, ablation, negative-control, trace, utility, theorem, proof-appendix, and reproducibility artifacts*

satisfying the equal-domain gate, then the bounded healthcare monitoring row satisfies the repository's equal artifact-discipline contract.

Proof. The gate requires the healthcare T11 fail-safe release runtime lemma and the healthcare T6 one-step fallback-validity lemma, then requires native runtime denominator rows for baselines, ablations, negative controls, utility, and reproducibility. The ORIUS row must have zero TSVR, valid runtime certificates, passing runtime witnesses, and useful work greater than the always-alert comparator. The result is an artifact-discipline guarantee for the promoted monitoring contract only; it does not assert regulated clinical deployment readiness. □

C.14 L1 — Reliability-Monotone Inflation

Lemma L1: Reliability-Monotone Inflation. Let $m_t = q_t/(w_t + \epsilon)$ with $q_t > 0$, $w_t \in [0, 1]$, and $\epsilon > 0$. Then m_t is monotonically nonincreasing in w_t.

Proof.

$$\frac{\partial m_t}{\partial w_t} = -\frac{q_t}{(w_t + \epsilon)^2} < 0.$$

Thus increasing reliability cannot increase the margin.

C.15 L2 — Safe-Set Antitonicity

Lemma L2: Safe-Set Antitonicity. If $X_1 \subseteq X_2$, then

$$\bigcap_{x \in X_2} C(x) \subseteq \bigcap_{x \in X_1} C(x).$$

Proof. Any action safe for every state in the larger set X_2 is safe for every state in its subset X_1. Hence the larger uncertainty set has a smaller or equal common safe-action set.

C.16 L3 — Intervention Threshold

Lemma L3: Intervention Threshold. If a candidate action a_t lies outside $\cap_{x \in X_t} C(x)$, any certified ORIUS release must repair, fallback, or deny release.

Proof. Membership in $\cap_{x \in X_t} C(x)$ is exactly the condition for the action to be safe for every state covered by the runtime uncertainty set. If the action is outside that set, releasing it

would violate the covered-release contract. Therefore a certified runtime must choose a repair, fallback, or denial path.

C.17 L4 — Observation-Ambiguity Safety Sandwich

Lemma L4: Observation-Ambiguity Safety Sandwich. Observation-only mandatory release inherits the lower-side ambiguity risks in T9 and T10. ORIUS covered release has upper-side risk bounded by its coverage miss probability under the T2/T3 covered-release contract.

Proof. The lower side follows by applying T9 to empty safe cores and T10 to two-state boundary ambiguity classes. For the upper side, ORIUS releases only actions safe over the covered state set. Thus violation can occur only when the true state is outside that covered set or an explicit runtime assumption is broken. Under the T2/T3 contract this event has probability no larger than the declared coverage miss probability. Combining the lower and upper statements gives the sandwich.

C.18 T11_Byzantine — Byzantine-Robust Reliability Aggregation

Theorem `T11_Byzantine`: Byzantine-Robust Reliability Aggregation. Suppose n telemetry channels emit scores in $[0,1]$, at most $b < n/2$ are Byzantine, and $\hat{w}_t$ is the b-trimmed mean after sorting all reported scores and removing the b smallest and b largest values. If the honest scores lie in an interval of radius ρ around w_t^H, then

$$|\hat{w}_t - w_t^H| \leq \rho.$$

Proof. Trimming removes the b smallest and b largest reported values. Since at most b reports are Byzantine, no value below all honest scores can remain after the lower trim: there are at most b such values, so they are among the b smallest reports. Similarly, no value above all honest scores can remain after the upper trim. All honest scores lie in $[w_t^H - \rho, w_t^H + \rho]$ by hypothesis. Hence every retained value that contributes to the trimmed mean lies in $[w_t^H - \rho, w_t^H + \rho]$. Their arithmetic mean lies in the same interval, which proves the bound. The theorem makes no claim for $b \geq n/2$ or for honest scores outside the stated interval.

C.19 T_stale_decay — Stale-Hold Uncertainty Growth

Theorem `T_stale_decay`: Stale-Hold Uncertainty Growth. Assume no fresh observation arrives for s steps and the latent state obeys $\|x^\star_{t+k} - x^\star_{t+k-1}\| \le L$ for $k = 1, \dots, s$. If the uncertainty radius at time t is r_t, then the conservative stale-hold radius is $r_{t+s} = r_t + Ls$.

Proof. By the triangle inequality,

$$\|x^\star_{t+s} - x^\star_t\| \le \sum_{k=1}^{s} \|x^\star_{t+k} - x^\star_{t+k-1}\| \le Ls.$$

Combining this drift radius with the existing uncertainty radius r_t gives $r_t + Ls$. Therefore any certified release at $t + s$ must be safe over the expanded set. This is a bounded-drift containment theorem, not a claim that a particular exponential decay schedule is physically derived.

C.20 T_minimax — Finite Ambiguity-Class Minimax Lower Bound

Theorem `T_minimax`: Finite Ambiguity-Class Minimax Lower Bound. Let $\mathcal{P}_\epsilon$ be a nonempty class of finite two-state boundary problems $P = (x_0, x_1, P_0, P_1, C)$ satisfying $d_{\mathrm{TV}}(P_0, P_1) \le \epsilon$ and $C(x_0) \cap C(x_1) = \varnothing$. For a mandatory observation-only controller π, define the worst-latent-state loss on P by

$$L(\pi, P) = \sup_{i \in \{0,1\}} \Pr_{O \sim P_i}[\pi(O) \notin C(x_i)].$$

Then

$$\inf_{\pi \in \Pi_{\mathrm{obs}}} \sup_{P \in \mathcal{P}_\epsilon} L(\pi, P) \ge \frac{1 - \epsilon}{2}.$$

Proof. Fix any π. For every $P \in \mathcal{P}_\epsilon$, the hypotheses of T10 hold by definition of the class, so

$$L(\pi, P) \ge \frac{1 - \epsilon}{2}.$$

Because the class is nonempty, taking the supremum over $P \in \mathcal{P}_\epsilon$ preserves the same lower bound for the fixed π. Taking the infimum over π then preserves it for the controller class. This is a finite ambiguity-class lower bound under the displayed loss; it is not a global minimax optimality theorem and does not assert a matching ORIUS upper bound.

C.21 T_sensor_converse — Sensor Necessity Under Adapter Semantics

Theorem `T_sensor_converse`: Sensor Necessity Under Adapter Semantics. Let x^j be a latent coordinate on which the safe-action map depends. Suppose the adapter observation map is invariant to that coordinate: whenever two states differ only in x^j, they have the same observation. If there exist x_0, x_1 differing only in x^j such that

$$h(x_0) = h(x_1) = o, \qquad C(x_0) \cap C(x_1) = \varnothing,$$

then no observation-only mandatory-release policy can certify safety over the ambiguity class containing x_0, x_1 without additional sensing, uncertainty expansion, fallback, or denial.

Proof. By the adapter-invariance hypothesis, x_0 and x_1 produce the same observation o, so both states lie in the nonempty ambiguity class $B(o)$. Because $C(x_0) \cap C(x_1) = \varnothing$, the common safe core over that class is empty. T9 then rules out certification by any total observation-only mandatory-release policy on $B(o)$. Additional sensing, uncertainty expansion, fallback, or denial are the escape routes because they change the observation map or leave the mandatory-release policy class.

C.22 T_trajectory_PAC — Finite-Horizon PAC Release Certificate

Theorem T_trajectory_PAC: Finite-Horizon PAC Release Certificate. Let Z_t be the true-state violation event at step t. Suppose

$$\Pr(Z_t = 1) \le \epsilon_t, \qquad t = 1, \dots, T.$$

Then

$$\Pr\left(\sum_{t=1}^{T} Z_t \ge 1\right) \le \sum_{t=1}^{T} \epsilon_t.$$

Therefore, if $\sum_{t=1}^{T} \epsilon_t \le \delta$, the trajectory is violation-free with probability at least $1 - \delta$. The same conclusion holds when the runtime supplies predictable conditional budgets $\Pr(Z_t = 1 \mid \mathcal{H}_{t-1}) \le \epsilon_t$, after taking expectations.

Proof. The event $\{\sum_{t=1}^{T} Z_t \geq 1\}$ is the union of the per-step violation events $\cup_{t=1}^{T}\{Z_t = 1\}$. By the union bound,

$$\Pr\left(\cup_{t=1}^{T}\{Z_t = 1\}\right) \leq \sum_{t=1}^{T} \Pr(Z_t = 1) \leq \sum_{t=1}^{T} \epsilon_t.$$

If the sum is at most δ, then the complementary no-violation event has probability at least $1 - \delta$. This defended surface is exactly the finite-horizon Bonferroni certificate; it does not claim a Ville or supermartingale strengthening and does not require temporal independence.

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